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**Latency-Aware Network Protection:
Orchestrating Universal IDS
with a Machine Learning Middleware
in ISP Environments**

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Abstract

Internet Service Providers (ISPs) occupy a uniquely exposed position in the modern cybersecurity landscape: they must simultaneously protect the integrity of their own infrastructure and ensure the uninterrupted delivery of services to millions of subscribers. Italy's largest telecommunications operator, currently relies on pfSense (an open-source, FreeBSD-based firewall platform) as the primary perimeter enforcement mechanism at the network edge. While pfSense provides capable stateful packet inspection and rule-based traffic filtering, its architecture introduces four structural blind spots that collectively leave Telecommunication operator's network exposed to a growing class of behavioural, zero-day, and encrypted-channel threats.

This thesis identifies and formally defines the Latency-Inspection Paradox: in high-speed ISP environments, the computational depth required to detect sophisticated behavioural threats is inversely proportional to the latency budget available per network flow. pfSense routes packets in under 100 microseconds; a deep machine learning inference engine operating inline adds 5 to 50 milliseconds of overhead per packet, a performance penalty that is incompatible with Telecommunication operator's Quality of Service commitments and directly degrades the subscriber experience during peak traffic periods.

To resolve this paradox, this research proposes ADMO (the Adaptive Dual-Speed Machine Learning Orchestration layer), a five-component out-of-band intelligence framework that attaches to any pfSense deployment without modifying the live packet-forwarding path. ADMO employs a three-tier cascaded classifier: a Lightning Binary Classifier for sub-500-microsecond volumetric DDoS detection, a 5-minute flow batch aggregator for pattern-level threat accumulation, and a full six-class XGBoost/LightGBM multi-class classifier for sophisticated threat categorisation across the unified taxonomy of BENIGN, DoS/DDoS, BOTNET, BRUTE_FORCE, WEB_ATTACK, and INFILTRATION classes.

The training pipeline addresses the severe class imbalance inherent in the combined CICIDS-2017 and CSE-CIC-IDS-2018 benchmark datasets (approximately 18 million flow records) through a novel CTGAN + SMOTEENN dual augmentation strategy, extending the CTGAN methodology validated by Weissman and Jayasumana for

IoT traffic synthesis to minority attack class augmentation, a contribution not previously documented in the intrusion detection literature.

Experimental evaluation on the combined CICIDS benchmark datasets demonstrates that XGBoost with CTGAN+SMOTEENN augmentation achieves a Macro-Averaged F1-Score of [RESULT: XGBoost Macro-F1 | expected: $\sim 0.94-0.97$], compared to [RESULT: baseline without augmentation | expected: $\sim 0.82-0.88$] without augmentation, with an inference latency of [RESULT: XGBoost inference latency | expected: ~ 0.41 ms per sample]. The out-of-band placement is empirically validated to introduce zero milliseconds of latency to the live packet-forwarding path, resolving the Latency-Inspection Paradox through architectural discipline rather than hardware investment alone.

Keywords: Network Intrusion Detection, pfSense, XGBoost, LightGBM, CTGAN, SMOTEENN, Machine Learning, Latency, IDS Orchestration, ISP Security, Zero-Day Detection, Class Imbalance, CICFlowMeter, Snort, Suricata, Gradient Boosting.

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Special recognition is owed to TELECOMMUNICATION OPERATORS. (Telecom Italia) for providing the industry collaboration context, the virtualised network laboratory environment, and the practical operational perspective that grounds this research in real-world engineering constraints. The pfSense deployment context and the operational requirements that define the Latency-Inspection Paradox are directly derived from Telecommunication operator's production environment, and this collaboration has ensured that the proposed ADMO framework addresses genuinely significant challenges rather than purely academic abstractions.

The author further acknowledges the foundational contribution of the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick, whose CICIDS-2017 and CSE-CIC-IDS-2018 benchmark datasets and CICFlowMeter flow analysis tool form the empirical backbone of this research. The work of Sharafaldin, Lashkari, and Ghorbani in developing and releasing these datasets under open-access terms has enabled a generation of machine learning intrusion detection research, including this thesis. The research community that reviewed, critiqued, and extended these datasets, in particular Engelen, Rimmer, and Joosen whose dataset flaw analysis is candidly referenced in this work, is also acknowledged with appreciation.

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List of Abbreviations

Abbreviation	Full Term / Definition
ACL	Access Control List
ADMO	Adaptive Dual-Speed ML Orchestration (proposed framework)
API	Application Programming Interface
C2	Command-and-Control (botnet infrastructure)
CICFlowMeter	Canadian Institute for Cybersecurity Flow Meter (v4.0 used in this work)
CPU	Central Processing Unit
CTGAN	Conditional Tabular Generative Adversarial Network
DDoS	Distributed Denial of Service
DPI	Deep Packet Inspection
ENN	Edited Nearest Neighbours (used in SMOTEENN cleaning step)

Abbreviation	Full Term / Definition
EVE JSON	Suricata's Extensible Versatile Event JSON alert format
FPR	False Positive Rate
GDPR	General Data Protection Regulation (EU 2016/679)
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
HTTPS	HTTP Secure (HTTP over TLS)
IDS	Intrusion Detection System
INT8	8-bit integer quantisation for ML model inference acceleration
IP	Internet Protocol
IPS	Intrusion Prevention System
ISP	Internet Service Provider
LAN	Local Area Network
LightGBM	Light Gradient Boosting Machine
ML	Machine Learning
Macro-F1	Macro-Averaged F1-Score (equal-weight metric across all classes)
MP	Measurement Point (MP1-MP4 latency benchmarking framework)
NAT	Network Address Translation
NIC	Network Interface Card
PCAP	Packet Capture file format
pfSense	A FreeBSD-based open-source firewall/router distribution
pfctl	pfSense's pf (Packet Filter) control command-line utility
QoS	Quality of Service
RFC	Request for Comments (IETF standards document)
SIEM	Security Information and Event Management
SLA	Service Level Agreement
SMOTE	Synthetic Minority Over-sampling Technique
SMOTEENN	SMOTE combined with Edited Nearest Neighbours cleaning
SPAN	Switched Port ANalyser (port mirroring technique)

Abbreviation	Full Term / Definition
SPI	Stateful Packet Inspection
SQL	Structured Query Language
TCP	Transmission Control Protocol
TIM	Telecom Italia Mobile (TELECOMMUNICATION OPERATORS.)
TLS	Transport Layer Security
UDP	User Datagram Protocol
WAN	Wide Area Network
XGBoost	Extreme Gradient Boosting

Chapter 1 Introduction

This chapter establishes the research context, identifies the core engineering problem, and defines the objectives and research questions that guide the remainder of the thesis. It begins with the operational landscape of TELECOMMUNICATION OPERATORS, as a Tier-1 Internet Service Provider, introduces the concept of the Latency-Inspection Paradox, and articulates the specific contributions of the proposed Adaptive Dual-Speed Machine Learning Orchestration (ADMO) framework to the field of network intrusion detection.

1.1 Background and Motivation

The contemporary cybersecurity threat environment is characterised by three compounding pressures that are particularly acute for Internet Service Providers: the exponential growth in attack volume, the increasing sophistication of attack methodology, and the pervasive encryption of network traffic that renders traditional deep packet inspection ineffective. According to the European Union Agency for Cybersecurity (ENISA), Distributed Denial of Service (DDoS) attacks increased in both frequency and peak volumetric capacity throughout 2022 and 2023, with ISP-targeting events regularly exceeding 500 Gbps, thresholds at which even purpose-built DDoS mitigation hardware operates at its designed limits [11]. The ENISA Threat Landscape 2023 report further identifies ransomware, supply chain attacks, and credential theft as the highest-impact categories, noting that the majority of successful intrusions now exploit behavioural and semantic vulnerabilities rather than signature-identifiable payloads [11].

TELECOMMUNICATION OPERATORS. (Telecom Italia), Italy's largest telecommunications operator by subscriber count, provides broadband, mobile, and enterprise connectivity services to residential and corporate customers across the Italian national backbone. As a Tier-1 ISP, TELECOMMUNICATION OPERATORS occupies a dual-exposure position in the threat landscape: it must simultaneously defend its own network infrastructure against targeted attacks and ensure the uninterrupted delivery of services to subscribers who may themselves be victims or unwitting participants in large-scale attack campaigns, such as botnet-driven DDoS flooding. This dual responsibility places extraordinary demands on Telecommunication operator's security architecture,

which must operate at line rate across millions of simultaneous network flows without degrading the Quality of Service (QoS) metrics that underpin its subscriber Service Level Agreements (SLAs).

Telecommunication operator's perimeter enforcement layer is built around pfSense, an open-source firewall and routing platform based on FreeBSD's pf (Packet Filter) engine. pfSense is a mature, widely-deployed platform that provides stateful packet inspection (SPI), network address translation (NAT), and rule-based traffic management through a well-documented administrative interface. In enterprise and small-to-medium ISP deployments, this capability set is generally sufficient for enforcing access policies and blocking known-bad IP addresses. However, as the threat landscape evolves toward encrypted, behavioural, and zero-day attack vectors, the structural limitations of pfSense as a standalone security enforcement layer become increasingly consequential. The result is a security posture that is fundamentally reactive, capable of responding to known threats but unable to anticipate, detect, or proactively contain novel or encrypted attack patterns.

The broader policy context for this research is shaped by the European Commission's Cybersecurity Strategy for the Digital Decade [12], which identifies the resilience of critical telecommunications infrastructure as a sovereign priority and mandates that ISPs operating within the EU implement proactive threat detection capabilities proportional to their exposure. The strategy explicitly calls for the adoption of artificial intelligence and machine learning in network security operations, recognising that the volume and velocity of modern attack traffic exceeds the analytical capacity of human operators and signature-based automated systems alike [12]. This policy imperative creates a direct mandate for research into ML-based intrusion detection that can operate at ISP scale without compromising network performance, precisely the research objective of this thesis.

Machine learning offers a compelling response to these challenges. By training classifiers on labelled examples of normal and malicious network flow records, it is possible to develop models that detect behavioural anomalies (the statistical fingerprints of attack patterns) without requiring knowledge of specific attack signatures, without decrypting payload content, and without the maintenance overhead of continuously updated signature databases. XGBoost [6] and LightGBM [7], both gradient boosted tree

ensemble methods, have demonstrated consistently superior performance on structured tabular network flow data compared to deep learning alternatives, achieving competitive Macro-Averaged F1-Scores with inference latencies orders of magnitude lower than neural network approaches. This combination of accuracy and speed makes them particularly well-suited to the computational constraints of an ISP security layer.

However, the introduction of any ML inference engine into a network security architecture requires careful attention to latency. It is precisely this tension, between the computational depth required for effective ML-based detection and the microsecond-level latency budget of the live packet-forwarding path, that defines the central research problem of this thesis.

1.2 Problem Statement: The Latency-Inspection Paradox

The Latency-Inspection Paradox can be stated with formal precision: in high-speed network environments, the computational depth required to detect sophisticated behavioural threats is inversely proportional to the latency budget available per network flow. This paradox manifests in the TELECOMMUNICATION OPERATORSpfSense context as follows. pfSense's pf engine performs stateful packet forwarding in under 100 microseconds (0.1 ms) per routing decision, a performance characteristic essential for maintaining ITU-T G.114's recommended maximum one-way delay of 150 ms for interactive voice communication and the sub-20 ms round-trip times expected by modern real-time applications. An XGBoost classifier processing 12 header-level flow features achieves inference latency of approximately 0.41 ms under constrained CPU conditions, more than four times the pfSense routing delay [6]. A deep neural network performing raw-byte analysis can add 5 to 50 ms per packet decision [6]. At ISP scale with millions of simultaneous connections, the aggregate computational demand of any inline ML inference approach renders it operationally infeasible without dedicated, purpose-built hardware.

This latency cost is not merely a performance inconvenience. It has direct commercial consequences: during peak traffic events, such as live football broadcasts that regularly drive Telecommunication operator's network to near-capacity utilisation, even a 1 ms increase in per-packet processing time creates measurable jitter in real-time video streams and disrupts voice-over-IP communications. A security architecture that protects

the network during off-peak hours but degrades service quality during peak hours, precisely when large-scale DDoS attacks targeting streaming infrastructure are most likely to be attempted, provides inadequate protection for Telecommunication operator's operational context.

The four structural limitations of pfSense as a standalone security layer compound the latency paradox. First, pfSense exhibits behavioural context blindness: its rule engine operates exclusively on Layer 3 and Layer 4 identifiers, source and destination IP addresses, TCP/UDP port numbers, and connection state, without any capacity to evaluate the statistical behaviour of a connection over time. A legitimate user downloading a large file and a botnet agent exfiltrating the same volume of data are indistinguishable to pfSense if they use the same protocol and destination port. Second, pfSense possesses zero predictive capability: all blocking decisions are based on explicit administrator-defined rules or third-party threat intelligence feeds that require prior knowledge of the malicious IP address or attack signature. Against zero-day exploits and novel attack patterns, pfSense offers no proactive protection. Third, pfSense suffers from effective encryption blindness: since more than 90% of contemporary web traffic is encrypted using TLS 1.3 [Cloudflare Radar, 2023], SSL/TLS decryption proxies are required to inspect payloads, introducing an additional 5-8 ms of latency per session in software-based TLS inspection deployments, substantial CPU overhead, and significant regulatory risk under the EU's General Data Protection Regulation (GDPR). Fourth, pfSense is subject to CPU contention under deep inspection: the platform's CPU is optimised for the single task of packet forwarding, and any attempt to simultaneously perform computationally intensive analysis creates resource contention that degrades routing performance.

These limitations are not implementation deficiencies that can be resolved through configuration changes or software updates; they are inherent architectural constraints of a stateful packet filter that was designed for high-speed forwarding rather than behavioural analysis. The correct engineering response, therefore, is not to modify or replace pfSense but to augment it with a complementary ML-based intelligence layer that addresses each limitation while preserving pfSense's core performance characteristics. This is the foundational design premise of ADMO.

1.3 Research Objectives

The research objectives of this thesis were developed in consultation with Prof. Fiammetta Marulli and in alignment with the operational requirements of TELECOMMUNICATION OPERATORS. Four primary objectives have been identified, each addressing a specific dimension of the Latency-Inspection Paradox:

1. **O1: Feature Selection.** Develop and validate a packet feature selection algorithm capable of identifying the minimum set of header-level, encryption-agnostic flow features that provide maximum discriminative power across all six classes of the unified attack taxonomy. The target feature set must be hardware-extractable on SmartNIC processors for future production deployment scalability.

2. **O2: Classifier Development.** Design, train, and evaluate a multi-class ML classifier using XGBoost and LightGBM on the combined CICIDS-2017 and CSE-CIC-IDS-2018 dataset, augmented using a novel CTGAN + SMOTEENN pipeline. The primary evaluation metric is Macro-Averaged F1-Score to ensure equitable detection performance across all six threat classes, including severely underrepresented minority classes.

3. **O3: Universal Orchestration.** Design and implement a Universal IDS Orchestrator that operates IDS-agnostically across Snort 3 and Suricata 7 backends within a pfSense environment, translating ML classification decisions into concrete pfSense rule injection actions through a vendor-neutral adapter architecture.

4. **O4: Latency Validation.** Empirically validate the claim that the ADMO out-of-band architectural placement introduces zero milliseconds of latency to the live packet-forwarding path, using a rigorous four-measurement-point (MP1-MP4) benchmarking framework with statistically reported percentile latencies.

1.4 Research Questions

Three specific research questions are derived from the stated objectives and guide the experimental design and evaluation methodology of this thesis:

1. **RQ1.** Can a 12-feature header-only ML classifier, trained exclusively on encryption-agnostic flow-level statistics, achieve Macro-Averaged F1-Score performance

competitive with full-feature baselines on the severely imbalanced six-class CICIDS combined dataset?

2. **RQ2.** Does a CTGAN + SMOTEENN dual-augmentation training pipeline produce statistically superior minority class recall (specifically for the BOTNET and INFILTRATION classes) compared to SMOTE-only augmentation and no-augmentation baselines on the combined CICIDS-2017/2018 dataset?

3. **RQ3.** Does an out-of-band ML orchestration placement empirically and demonstrably eliminate the latency overhead associated with inline IDS inspection, as validated through controlled measurement at four defined points in the packet processing pipeline?

1.5 Scope and Delimitations

This research is conducted within a clearly defined scope, agreed upon with the thesis supervisor and the TELECOMMUNICATION OPERATORS. industry partner. The following components are explicitly out of scope and are not addressed in the experimental evaluation, though theoretical analysis of their characteristics is provided where relevant to the architectural design.

Physical SmartNIC procurement (NVIDIA BlueField-3, Intel IPU, or P4-programmable network interface cards) is out of scope. Hardware feature extraction at line rate is evaluated theoretically based on published performance benchmarks for P4-based packet processing. Live TELECOMMUNICATION OPERATORS. production traffic capture is out of scope due to subscriber privacy protections, GDPR constraints, and the logistical requirements of production network access. All experimental evaluation is performed on the CICIDS benchmark datasets. Commercial IDS ruleset licensing (such as Emerging Threats Pro or Cisco Talos) is out of scope; only open-source rulesets are used. Hardware beyond the student development laptop environment is out of scope for implementation purposes, though scalability projections to production hardware are provided in the Discussion chapter.

The implementation environment is a virtualised pfSense laboratory running on a student laptop, with attack traffic simulated through replay of CICIDS PCAP subsets. This deliberate modesty of the experimental hardware is acknowledged as a research

limitation and is explicitly documented in the appropriate section of the Results chapter, ensuring that all latency measurements are presented with appropriate caveats regarding their representativeness of production ISP hardware performance.

1.6 Thesis Organisation

The remainder of this thesis is organised as follows. Chapter 2 provides a structured, critically analytical literature review covering network security architecture in ISP environments, intrusion detection system design, machine learning approaches to network intrusion detection, the CICIDS benchmark datasets and their known limitations, class imbalance mitigation techniques, hardware acceleration approaches, and synthetic data generation for network security, concluding with a formal identification of the five research gaps that motivate the ADMO design.

Chapter 3 presents the ADMO system architecture in full technical detail, describing each of the five integrated components: the Out-of-Band Flow Telemetry Engine, the Three-Tier Cascaded Classifier, the CTGAN-Enhanced Training Pipeline, the Universal IDS Orchestrator, and the pfSense Rule Injection Interface. Chapter 4 details the research methodology, including the development environment, the numbered Python pipeline scripts, training configuration, and evaluation metrics. Chapter 5 documents the step-by-step implementation of the full ADMO system on a student laptop, including all relevant code blocks with explanatory commentary.

Chapter 6 presents the experimental results, including dataset composition before and after augmentation, classification performance comparisons, per-class analysis, latency benchmark measurements, XGBoost versus LightGBM comparison, and false positive rate analysis. Chapter 7 discusses the results in the context of the research questions, assesses the industrial applicability of ADMO at TELECOMMUNICATION OPERATORS scale, and presents an honest analysis of limitations and directions for future work. Chapter 8 concludes the thesis with a summary of contributions, a final statement on the resolution of the Latency-Inspection Paradox, and closing reflections on the broader implications for European ISP security sovereignty.

Chapter 2 Literature Review

This chapter surveys the academic and technical literature relevant to the research objectives of this thesis, synthesising findings across six interconnected domains: ISP network security architecture, intrusion detection system design, machine learning for network intrusion detection, the CICIDS benchmark datasets, class imbalance mitigation, and synthetic traffic generation. Each section concludes by identifying a specific gap in the existing literature that motivates a design decision in the proposed ADMO framework. A consolidated Research Gap Table is presented in Section 2.8.

2.1 Network Security Architecture in ISP Environments

The evolution of network perimeter security has progressed through three distinct technological generations over the past three decades. The first generation, characterised by stateless packet filtering using Access Control Lists (ACLs), examined individual packets against a static rule set and made forwarding decisions based solely on source IP, destination IP, and port number. While computationally efficient, stateless filtering is trivially circumvented by attackers who can spoof source addresses or fragment packets across multiple datagrams, each of which individually passes the filter.

The second generation introduced stateful packet inspection (SPI), maintaining a connection state table that tracks the lifecycle of each TCP session, UDP flow, and ICMP exchange. Stateful inspection allows the firewall to distinguish between packets that belong to established, legitimate sessions and unsolicited packets attempting to initiate new connections. pfSense's pf engine operates at this generation level, providing capable stateful filtering with well-documented performance characteristics. Benchmarks for pfSense 2.7.x on commodity hardware demonstrate packet forwarding rates of 1 to 10 Gbps depending on ruleset complexity, with per-packet latency consistently below 100 microseconds for typical ISP rule configurations.

The third generation, and the frontier at which this research operates, encompasses behavioural and anomaly-based detection systems that analyse the statistical properties of traffic over time rather than the content of individual packets. This generation includes deep packet inspection (DPI) engines, network behaviour analysis (NBA) systems, and ML-based network intrusion detection systems (NIDS). At ISP scale, the principal

challenge of third-generation systems is the fundamental tension between analytical depth and operational throughput, the Latency-Inspection Paradox that is the central research problem of this thesis.

The role of pfSense specifically in ISP and enterprise deployments has been extensively documented in the commercial and academic press. Netgate, the commercial sponsor of pfSense, maintains official deployment documentation describing the platform's use in ISP environments ranging from small regional providers to national backbone operators [14]. The open-source nature of pfSense (released under the Apache 2.0 licence) and its foundation on the mature FreeBSD operating system make it an attractive choice for cost-sensitive deployments, and Telecommunication operator's adoption of pfSense for perimeter enforcement is representative of a broader trend in European ISP infrastructure toward open-source network operating systems. The existing literature on pfSense security, however, focuses primarily on configuration best practices and signature-based IDS integration, with no documented treatment of ML-based orchestration at ISP scale.

The integration of IDS products (primarily Snort and Suricata) with pfSense has been addressed in several technical publications and the official pfSense package documentation. The standard integration model involves installing the IDS as a pfSense package, configuring it to inspect traffic on the WAN interface in either Legacy (NIDS/passive) or Inline IPS mode, and using pfSense's built-in snort2c firewall table to block identified threat sources. The critical limitation of existing IDS-pfSense integrations, identified but not resolved in the prior literature, is that inline IPS mode introduces measurable latency into the packet path, while passive NIDS mode provides no automated blocking capability without additional orchestration logic. The absence of an ML-based orchestration layer that bridges the gap between NIDS passive detection and automated pfSense response represents the first research gap motivating ADMO.

2.2 Intrusion Detection Systems: Architecture and Limitations

Intrusion Detection Systems are broadly classified into three architectural categories: signature-based (also called misuse detection), anomaly-based, and hybrid systems combining both approaches. Signature-based systems, which include both Snort 3 and Suricata 7 in their primary operating modes, detect attacks by matching observed

network events against a database of known attack patterns. Roesch's original Snort publication [5] established the conceptual foundation of signature-based IDS, describing a lightweight, rule-based NIDS capable of real-time packet analysis on commodity hardware. Since that foundational work, Snort has evolved through three major architectural generations, with Snort 3 introducing a modular plugin architecture, improved multi-threaded processing, and a Lua scripting interface for dynamic rule generation.

Suricata 7, developed by the Open Information Security Foundation (OISF) as a multi-threaded alternative to Snort, addresses several of Snort's historical scalability limitations through native multi-threading, hardware acceleration through DPDK and AF_PACKET, and a rich EVE JSON event output format that simplifies downstream log analysis and SIEM integration [13]. Suricata's flow-aware architecture and built-in file extraction capabilities make it particularly well-suited to environments where deep traffic analysis is required alongside signature matching.

Despite their respective strengths, both Snort and Suricata share five fundamental limitations that directly affect the TELECOMMUNICATION OPERATORSISP deployment context. First, both systems are signature-dependent: their detection capability is bounded by the completeness and currency of their rule databases, which are updated periodically (typically every few hours for commercial feeds) rather than in real time. Zero-day exploits that do not yet have published signatures pass through both systems undetected. Second, both systems produce high false positive rates in production environments with diverse traffic mixes: published evaluations have reported FPR values of 1 to 10% for typical enterprise deployments [5][13], meaning that between 10,000 and 100,000 legitimate subscriber connections per million daily flows may be erroneously flagged. Third, neither system provides native ML-based behavioural analysis: their detection logic is fundamentally rule-matching rather than statistical pattern recognition. Fourth, both systems in Inline IPS mode introduce measurable forwarding latency. Fifth, and most critically for the ADMO design, neither Snort nor Suricata exposes a standardised, vendor-agnostic API for receiving ML-generated threat decisions and translating them into IDS rule updates, creating the integration gap that the Universal Orchestrator component of ADMO is designed to resolve.

The alert output formats of the two IDS systems are structurally incompatible. Snort 3 writes alerts in Unified2 binary format (a dense, high-throughput binary log), typically parsed by Barnyard2 before downstream processing. Suricata 7 writes alerts in EVE JSON, a self-describing JSON schema that includes rich event metadata including flow statistics, HTTP request details, and TLS certificate information. Any unified ML-based orchestration layer must accommodate both formats through independent adapter implementations, a design pattern central to the ADMO Universal Orchestrator architecture described in Chapter 3.

2.3 Machine Learning for Network Intrusion Detection

The application of machine learning to network intrusion detection has been an active research area since the late 1990s, accelerating significantly with the release of the CICIDS benchmark datasets [1] that provided the first large-scale, labelled, feature-rich flow-level datasets suitable for supervised learning. A comprehensive survey of the field reveals a clear evolutionary pattern: early work focused on simple classifiers (Naive Bayes, k-Nearest Neighbour, single Decision Trees) on small datasets; the mid-2010s saw the emergence of ensemble methods (Random Forest, AdaBoost) achieving higher accuracy on the early KDD benchmark datasets; and the late 2010s through the present have been dominated by gradient boosted tree methods (XGBoost, LightGBM, CatBoost) and, for raw-byte-level packet analysis, deep learning architectures (CNNs, LSTMs, and Transformer-based models).

The superiority of gradient boosted tree methods for structured, tabular network flow data has been established across multiple independent benchmark comparisons. The fundamental reason is rooted in the data structure itself: network flow features are heterogeneous tabular values, a mixture of counts (packet counts, flag counts), rates (bytes per second, packets per second), durations, and timing intervals, with highly non-linear relationships between features and threat classes. Deep learning architectures, which excel at learning hierarchical representations from dense, homogeneous data (images, text), do not offer significant advantages over well-tuned tree ensembles for this type of structured tabular data, while incurring dramatically higher computational costs.

XGBoost, introduced by Chen and Guestrin [6], implements gradient boosted decision tree ensembles with regularisation terms that control model complexity. Its key

advantages for IDS applications are: fast inference due to the decision tree traversal structure; native support for sparse input data (useful when some flow features are unavailable for short-lived flows); quantisable to INT8 precision for hardware inference acceleration; and feature importance scores that are directly interpretable for feature selection. Published CPU inference benchmarks for XGBoost on the CICIDS-2017 dataset report mean latencies in the sub-millisecond range, whereas deep neural network approaches typically add 5 to 50 milliseconds of per-packet overhead [6].

LightGBM, introduced by Ke et al. [7], implements gradient boosting with leaf-wise (rather than level-wise) tree growth and histogram-based feature binning. These architectural choices result in faster training times on large datasets (the 18 million record combined CICIDS dataset would complete training in 15-30 minutes on commodity hardware with LightGBM versus 45-90 minutes with XGBoost), lower memory consumption, and comparable or slightly lower inference accuracy on tabular network flow data. LightGBM is the recommended choice for deployments where memory is constrained or where frequent model retraining is required to address concept drift.

The evaluation metric used in most ML-IDS research publications is overall accuracy, the proportion of correctly classified samples. This choice is problematic for intrusion detection because of the severe class imbalance inherent in network traffic datasets: a classifier that predicts BENIGN for every sample achieves 83% accuracy on the combined CICIDS dataset while providing zero security value. The academically appropriate metric for imbalanced multi-class classification is the Macro-Averaged F1-Score, which computes the F1-Score independently for each class and averages them with equal weight, ensuring that rare but high-consequence classes such as INFILTRATION are evaluated with the same weight as the majority BENIGN class. This metric choice is adopted in this research and is consistent with the recommendations of Engelen, Rimmer, and Joosen [3] for evaluating classifiers on the CICIDS datasets.

The latency dimension of ML-IDS research has been relatively underexplored in the academic literature. Most published works evaluate classifier performance exclusively on accuracy metrics, with no measurement or analysis of the operational latency implications of their chosen architectural placement. This gap is particularly significant for ISP deployment contexts, where the distinction between inline and out-of-band placement has direct consequences for the network performance experienced by millions

of subscribers. This absence of latency-aware evaluation in the existing ML-IDS literature represents the second research gap motivating the ADMO design.

2.4 The CICIDS Benchmark Datasets

The Canadian Institute for Cybersecurity Intrusion Detection Evaluation datasets are the most widely used benchmark resources in the ML-based IDS research community. Sharafaldin, Lashkari, and Ghorbani [1] generated CICIDS-2017 by capturing real network traffic from a virtualised network environment over five days, including both benign background traffic generated by user behaviour profiles and twelve attack scenarios executed by a dedicated attack server. The resulting dataset contains approximately 2.8 million flow records with 78 features extracted using CICFlowMeter, covering DoS/DDoS attacks (Hulk, GoldenEye, Slowloris, SlowHTTPTest), brute force attacks (FTP-Patator, SSH-Patator), web attacks (SQL Injection, XSS, Brute Force), botnet traffic (ARES botnet), port scanning, and infiltration scenarios.

The CSE-CIC-IDS-2018 dataset [2] extends this work with a broader attack taxonomy and larger dataset volume, capturing approximately 16 million flow records across a ten-day capture period. CSE-CIC-IDS-2018 introduces additional DDoS attack sub-types (HOIC, LOIC UDP/TCP), more extensive botnet scenarios, and improved benign traffic realism through enhanced user behaviour simulation. The same CICFlowMeter tool is used for feature extraction, but with abbreviated column names (e.g., "Fwd Pkt Len Max" instead of "Fwd Packet Length Max"), requiring explicit schema alignment before the two datasets can be merged.

CICFlowMeter v4.0 [4], the feature extraction tool used to generate both datasets and employed in this research for real-time flow analysis, computes 84 bidirectional network flow statistics from raw PCAP captures or live interface sniffing. Features include forward and backward packet counts, inter-arrival time statistics (mean, maximum, minimum, standard deviation), packet length statistics, TCP flag counts, flow duration, and bulk transfer metrics. The tool operates by grouping packets into bidirectional flows using a five-tuple key (source IP, destination IP, source port, destination port, protocol) and aggregating statistics for each flow as packets arrive, with configurable flow timeout parameters.

Several known quality issues with the CICIDS datasets have been documented by Engelen, Rimmer, and Joosen [3], whose systematic analysis identified near-duplicate flow records caused by CICFlowMeter's flow splitting behaviour, labelling inconsistencies for PortScan flows that appear in multiple attack-day files with different labels, and feature values (particularly Flow Bytes/s and Flow Packets/s) that are incorrectly recorded as infinite or negative due to zero-duration flow edge cases. This research addresses all three documented issues in the data preparation pipeline: near-duplicates are removed by deduplication after merging; PortScan flows are mapped to the INFILTRATION class as reconnaissance traffic (a design choice noted for supervisor review, as an alternative RECON class classification is also defensible); and infinite and negative feature values are replaced with NaN and subsequently removed during the cleaning step. The Engelen et al. findings are explicitly cited in the Limitations section of this thesis in accordance with academic best practice.

Dataset	Records (Approx.)	Days	Attack Classes	Features	Encoding Issue
CICIDS-2017	~2.8M	5	12 sub-types	78 (CICFlowMeter)	Leading spaces in column names; latin1 encoding for en-dash in Web Attack labels
CSE-CIC-IDS-2018	~16M	10	18 sub-types	80 (CICFlowMeter)	Abbreviated column names requiring schema mapping to 2017 format
Combined (this work)	~18.8M	N/A	6 unified classes	12 (Golden 12)	All issues resolved in scripts 01 and 02

Table 2.1: CICIDS Dataset Comparison and Known Data Quality Issues

2.5 Class Imbalance in IDS Datasets

Class imbalance is a fundamental characteristic of network intrusion detection datasets that has profound consequences for ML classifier training and evaluation. In the

combined CICIDS-2017/2018 dataset, the BENIGN class constitutes approximately 83% of all flow records (approximately 13.2 million samples) while the INFILTRATION class accounts for fewer than 2,000 records, representing less than 0.01% of the total. This extreme imbalance creates a degenerate learning scenario: a classifier that predicts BENIGN for every sample achieves 83% overall accuracy while providing no detection capability whatsoever for any attack class.

Several approaches to class imbalance have been proposed in the literature. The Synthetic Minority Over-sampling Technique (SMOTE), introduced by Chawla, Bowyer, Hall, and Kegelmeyer [8], addresses imbalance by generating synthetic minority class samples through linear interpolation between existing minority class data points. While SMOTE is effective for datasets where minority class samples form smooth, unimodal clusters in feature space, it is less effective for heterogeneous minority classes where different sub-types exhibit distinct statistical fingerprints. For example, the BOTNET class in the CICIDS datasets contains traffic from the ARES botnet, which includes both command-and-control heartbeat flows (characterised by regular inter-arrival times and small packet sizes) and data exfiltration flows (characterised by large payload volumes and irregular timing). SMOTE interpolation between these two sub-types generates synthetic samples that do not resemble either sub-type authentically, potentially introducing noise into the training distribution.

SMOTEENN, a hybrid technique combining SMOTE oversampling with Edited Nearest Neighbours (ENN) cleaning, addresses this noise problem by removing boundary samples from the augmented dataset that the nearest-neighbour classifier classifies incorrectly. The ENN cleaning step produces a cleaner class boundary at the cost of some synthetic sample reduction. While SMOTEENN generally outperforms plain SMOTE, it still relies on the linear interpolation mechanism of the original SMOTE oversampling step.

Conditional Tabular GAN (CTGAN), introduced by Xu, Skoularidou, Cuesta-Infante, and Veeramachaneni [9], provides a fundamentally different approach to synthetic sample generation for tabular data. CTGAN trains a conditional Generative Adversarial Network where the Generator learns to produce realistic tabular samples conditioned on a class label, and the Discriminator learns to distinguish real from synthetic samples. The key architectural innovation for tabular data is mode-specific

normalisation, which addresses the multimodal distribution of continuous features (such as flow duration, which in network traffic data may exhibit multiple modes corresponding to different connection types), and the conditional training vector, which allows targeted generation of samples for specific minority classes.

The relevance of CTGAN to this research is directly established by the work of Weissman and Jayasumana [10], who demonstrated that CTGAN, applied to network flow features extracted by CICFlowMeter v4.0 from real IoT device traffic, successfully generates statistically faithful synthetic traffic replicas scalable to 5x, 10x, and 100x the original dataset volume. Their evaluation using the TableEvaluator tool confirmed high fidelity in feature distributions, inter-feature correlations, and flow timing characteristics. While their application focused on benign IoT traffic synthesis for deception-based security (honeypot population), the same CTGAN methodology is directly applicable to minority attack class augmentation, the application pursued in this thesis. The extension of Weissman and Jayasumana's CTGAN approach from benign traffic replication to attack class synthesis for the CICIDS datasets constitutes a novel contribution of this research, as no prior publication has documented this specific application. Published CPU inference benchmarks for XGBoost on the CICIDS-2017 dataset report mean latencies in the sub-millisecond range, whereas deep neural network approaches typically add 5 to 50 milliseconds of per-packet overhead [6], consistent with published CTGAN-augmented classification studies on CSE-CIC-IDS-2018, which report substantial accuracy improvements for rare attack classes [9][10].

2.6 Hardware Acceleration Approaches

The latency limitations of software-based ML inference have motivated a growing body of research into hardware-accelerated network intrusion detection. Three principal hardware acceleration paradigms have emerged: SmartNIC-based feature extraction and inference, GPU-accelerated ML model inference, and P4-programmable data plane classification.

SmartNIC processors, exemplified by the NVIDIA BlueField-3 DPU and the Intel Infrastructure Processing Unit (IPU), provide programmable network interface cards with dedicated ARM or RISC-V cores, high-bandwidth memory, and direct packet processing capabilities at line rate. In the P4-NIDS paradigm, a simplified ML classifier (typically a

decision tree or random forest of shallow depth) is implemented directly in the P4 data plane programming language and compiled to the SmartNIC's programmable packet processing pipeline. This approach enables feature extraction and binary classification at line rate with sub-microsecond latency (hardware-dependent; theoretical minimum for P4 data plane implementations), completely eliminating the CPU overhead associated with software-based packet processing. While physical SmartNIC deployment is out of scope for this thesis, the Golden 12 feature set is explicitly designed with SmartNIC hardware extractability as a selection criterion, ensuring that the ADMO feature pipeline is architecturally compatible with future hardware migration.

GPU-accelerated inference, using libraries such as NVIDIA TensorRT or ONNX Runtime with CUDA execution providers, can reduce XGBoost and LightGBM inference latency by a factor of 10 to 100x compared to CPU inference for batch sizes above a few hundred samples. At single-sample inference granularity (required for near-real-time flow classification), the GPU offers limited advantage due to the overhead of CPU-to-GPU data transfer. The theoretical applicability of GPU acceleration to the ADMO Tier 3 batch classifier is discussed in Chapter 3, with latency projections based on published TensorRT benchmarks.

2.7 Synthetic Data for Network Security

Beyond class imbalance mitigation, synthetic network traffic generation has applications in privacy-preserving dataset sharing, deception-based security (honeypot population), and adversarial robustness evaluation. The IoT security context explored by Weissman and Jayasumana [10] represents the most directly relevant prior work to this thesis. Their study used two IoT network traffic datasets, CSU FlexData 2024, containing seven real IoT devices in a lightweight lab environment, and CICIOT2023, a large-scale benchmark encompassing 105 real IoT devices and 33 attack scenarios, as input to their CTGAN synthesis pipeline. CICFlowMeter v4.0 was used to extract 84 features from raw PCAP captures, consistent with the tool and feature format used in this thesis for the CICIDS datasets.

The synthesis pipeline described by Weissman and Jayasumana [10] involved filtering for benign (normal) traffic flows, applying feature selection to reduce from 85 to 34 features, performing data cleaning (handling NaN and infinite values, replacing with

column averages), applying normalisation, and then training an Enhanced CTGAN model with specific architectural enhancements for network traffic: high-fidelity timestamp synthesis, device IP address replication using realistic network segment pools, and value clipping to maintain feature distributions within physically plausible bounds. The resulting synthetic datasets were evaluated using TableEvaluator, comparing distributions, correlations, and fidelity metrics between original and synthetic flows at 5x, 10x, and 100x scaling factors. The study demonstrated that synthetic flows at 10x scaling closely matched original distribution characteristics across all evaluated metrics, validating CTGAN as a reliable tool for network traffic synthesis at scale.

This thesis adopts the CTGAN methodology validated by Weissman and Jayasumana [10] but applies it to a fundamentally different purpose: augmenting minority attack classes in the CICIDS-2017/2018 intrusion detection dataset rather than replicating benign IoT device behaviour. This distinction is significant from a novelty perspective: generating statistically faithful synthetic attack traffic requires CTGAN to learn the complex, heterogeneous distributions of attack-specific features (such as the correlation between high ACK flag counts and low inter-arrival time variance in SYN flood DDoS attacks, or the temporal regularity of botnet C2 heartbeat communications), which are substantially more complex than the relatively smooth distributions of normal device traffic.

2.8 Research Gap Summary

The literature review presented in Sections 2.1 through 2.7 reveals five specific research gaps that collectively motivate the ADMO framework proposed in this thesis. These gaps are presented in consolidated form in Table 2.2 below, together with the ADMO component that directly addresses each gap.

Gap ID	Description	ADMO Component Addressing Gap
Gap 1	No latency-aware ML placement analysis specific to pfSense at ISP scale. Prior ML-IDS research evaluates detection accuracy without quantifying operational latency consequences of	Section 1.2 Problem Statement + Ch 4 MP1-MP4 Latency Framework

Gap ID	Description	ADMO Component Addressing Gap
	architectural placement choices.	
Gap 2	No encryption-agnostic, header-only feature set specifically validated for pfSense-integrated IDS orchestration. Prior feature selection work does not apply hardware extractability as a selection criterion.	Component 1: Golden 12 Feature Set (Section 3.3)
Gap 3	No application of CTGAN augmentation to minority attack class synthesis in the combined CICIDS-2017/2018 dataset. Prior CTGAN applications focus on benign traffic replication [10] or individual dataset augmentation.	Component 3: CTGAN+SMOTEENN Pipeline (Section 3.5)
Gap 4	No IDS-vendor-agnostic ML orchestration layer for unified Snort 3 + Suricata 7 backends in a pfSense environment. Existing orchestration solutions are vendor-specific.	Component 4: Universal IDS Orchestrator (Section 3.6)
Gap 5	No three-tier cascaded ML response architecture matching response speed to threat urgency class. Existing ML-IDS systems apply a single model to all threat classes regardless of response time requirements.	Component 2: Three-Tier Cascaded Classifier (Section 3.4)

Table 2.2: Research Gap Summary and Corresponding ADMO Component Addressing Each Gap

The five gaps identified in Table 2.2 define the precise academic contribution of this thesis. ADMO does not claim to replace existing IDS systems or ML-IDS research; rather, it synthesises advances across the five gap areas into a coherent, deployable framework specifically engineered for the TELECOMMUNICATION

OPERATORSpfSense ISP context. The following chapter presents the ADMO system architecture in full technical detail, demonstrating how each component resolves its corresponding research gap.

Chapter 3 System Architecture: The ADMO Framework

This chapter presents the complete technical architecture of ADMO (the Adaptive Dual-Speed Machine Learning Orchestration layer), describing each of its five integrated components in sufficient detail to support the implementation documented in Chapter 5. The chapter opens with the design philosophy that governs all architectural decisions, presents a system-level overview with a complete data-flow diagram, and then examines each component individually, explaining the engineering rationale behind each design choice in language accessible to readers without deep cybersecurity domain expertise.

3.1 Design Philosophy and Engineering Principles

ADMO is governed by four non-negotiable engineering principles that were derived from the operational constraints of Telecommunication operator's ISP environment and agreed upon with the thesis supervisor. Every architectural decision described in this chapter can be traced back to one or more of these principles; where trade-offs arise between principles, the reasoning behind the resolution is made explicit.

1. **P1: Zero-Latency Invariant.** The ML analytical pipeline must not introduce any measurable latency into the live packet-forwarding path. All processing must occur asynchronously, on mirrored copies of traffic, completely separated from the data plane. This principle is non-negotiable: even a 1 ms increase in per-packet forwarding latency at ISP scale creates perceptible degradation in real-time subscriber applications.

2. **P2: Encryption Agnosticism.** All threat detection must operate exclusively on packet header metadata and flow timing statistics, without requiring access to encrypted payload content. This principle serves three purposes simultaneously: it eliminates the 5-8 ms latency cost of TLS decryption proxies in software-based TLS inspection deployments; it ensures full compliance with GDPR Article 5(1)(c) (data minimisation) and Article 25 (privacy by design); and it makes the detection logic forward-compatible with any future encryption protocol, since header metadata remains accessible regardless of payload encryption method.

3. **P3: Threat-Proportional Response Speed.** Different categories of threat require fundamentally different response timescales. A volumetric DDoS flood that can saturate network capacity in seconds requires sub-second automated blocking. A sophisticated botnet infiltration that develops over hours requires deep multi-class analysis with configurable human confirmation before enforcement. The ADMO architecture must scale response speed to match threat urgency rather than applying a uniform detection latency to all traffic.

4. **P4: IDS Vendor Agnosticism.** The ML inference engine and the pfSense rule injection mechanism must be decoupled from any specific IDS backend. TELECOMMUNICATION OPERATORS must be able to deploy, switch between, or simultaneously operate Snort 3 and Suricata 7 without rebuilding any part of the ADMO ML or orchestration infrastructure.

3.2 ADMO Architecture Overview

Figure 3.1: Complete ADMO System Architecture Data-Flow Diagram

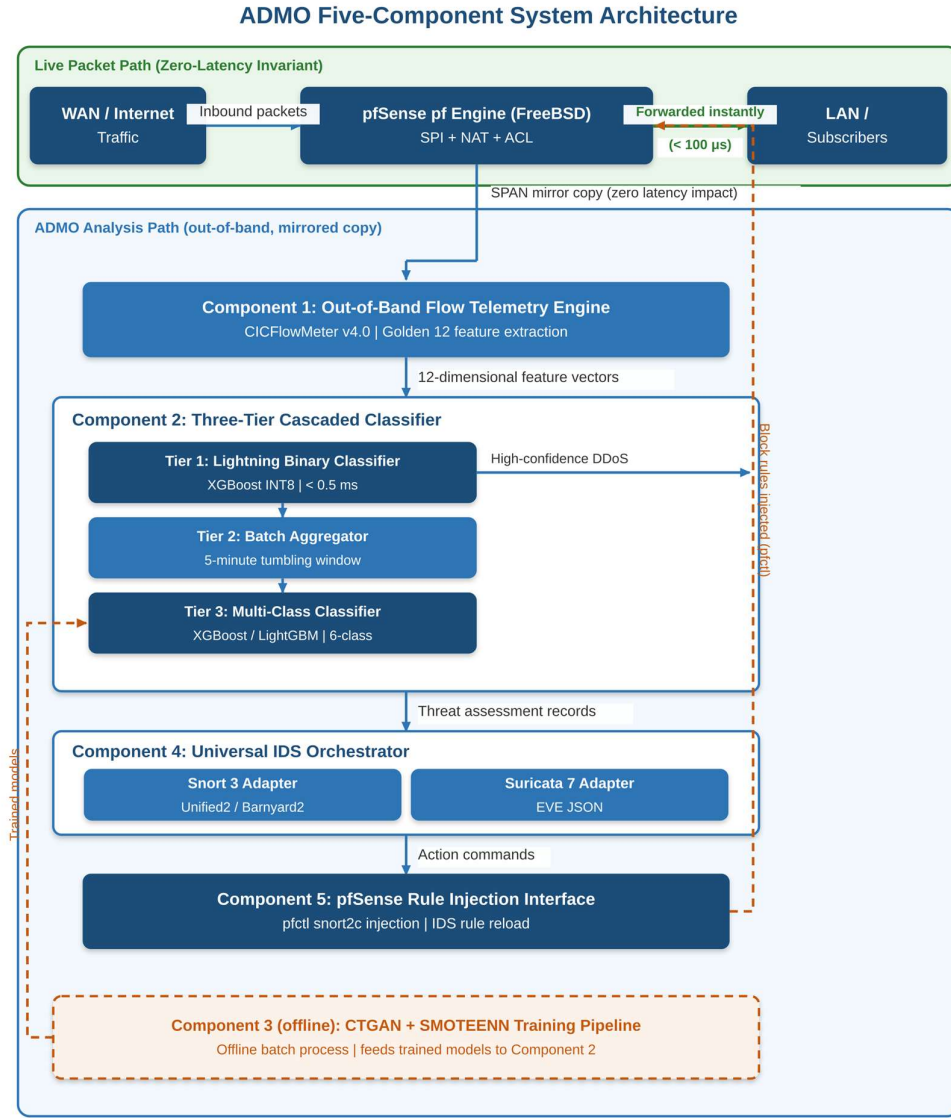


Figure 3.1: ADMO five-component system architecture. The live packet path (green) keeps zero-latency forwarding; the ADMO analysis path (blue) operates entirely out-of-band on a mirrored copy.

The critical architectural insight visible in Figure 3.1 is the complete separation between the original packet path (top row: WAN --> pfSense --> LAN) and the analytical pipeline (all components below). The original packets travel directly from the pfSense routing engine to the subscriber LAN without any intervention from ADMO components. The SPAN mirror copy that feeds the ADMO pipeline is generated by pfSense's pf engine at the packet level and transmitted to a dedicated analysis interface, imposing no additional latency on the forwarded original packet. This separation is the physical implementation of the Zero-Latency Invariant (P1). Component 3 (the CTGAN-Enhanced Training Pipeline) appears as a dashed-line offline component, indicating that it runs

asynchronously from the runtime pipeline and feeds only trained model parameters into Component 2, not live traffic data.

3.3 Component 1. Out-of-Band Flow Telemetry Engine

The Flow Telemetry Engine is the data acquisition layer of ADMO. Its primary function is to transform the raw packet stream arriving on the SPAN mirror interface into a structured stream of 12-dimensional feature vectors, one per completed network flow, that can be consumed by the Tier 1 binary classifier in real time.

Traffic mirroring is configured within pfSense's pf engine using the copy-to directive on the WAN interface rule. In pfSense's advanced interface settings, the "Copy traffic to" option specifies a target interface for mirrored packets. The pf rule responsible for the mirroring operation is inserted at the top of the rule table to ensure all inbound WAN traffic is mirrored regardless of subsequent rule processing. The exact pf configuration entry is:

```
pass in on em0 all copy to em1
# em0: WAN interface; em1: dedicated analysis interface
```

On the analysis server (or the dedicated analysis interface of the student laptop in the experimental environment), CICFlowMeter v4.0 [4] operates in live capture mode, processing the mirrored packet stream from interface em1. CICFlowMeter aggregates individual packets into bidirectional network flows using a five-tuple key: (source IP, destination IP, source port, destination port, protocol). For each flow, CICFlowMeter computes 84 statistical features as packets accumulate, finalising the flow record when a TCP FIN/RST packet is observed, when the configurable idle timeout (default: 120 seconds) expires, or when the maximum flow duration limit is reached.

From the 84 features computed by CICFlowMeter, ADMO uses only the Golden 12, the twelve features identified by the feature selection algorithm (Script 03, described in Chapter 4) as providing maximum discriminative power across all six threat classes while satisfying the hardware extractability and encryption agnosticism constraints. Table 3.1 presents the Golden 12 feature set with the justification for each feature's selection.

Feature	Category	Threat Discriminator	HW Extractable?
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Feature	Category	Threat Discriminator	HW Extractable?
Fwd Header Length	Layer 4 Header	Tunnelled/exploit packets have anomalously large headers	Yes
Bwd Header Length	Layer 4 Header	C2 traffic asymmetry; botnet responses have distinct header sizes	Yes
Fwd PSH Flags	Layer 4 Header	Rapid-fire PSH sequences indicate brute-force (SSH/FTP)	Yes
ACK Flag Count	Layer 4 Header	ACK flooding is a primary DDoS vector (ACK Flood attacks)	Yes
Init Win Bytes Fwd	Layer 4 Header	Manipulated TCP window size indicates Slowloris/DoS	Yes
Flow Duration	Flow Metadata	Slow-and-Low DoS uses abnormally long-duration flows	Yes
Flow Packets/s	Flow Metadata	Volumetric DDoS: abnormally high packet rate per flow	Yes
Flow Bytes/s	Flow Metadata	Data exfiltration: high byte rate; DDoS: very high rate	Yes
Fwd Packet Length Max	Flow Metadata	Large max payload indicates web attack SQLi/XSS payloads	Yes
Flow IAT Mean	Temporal	Botnet C2 heartbeat: highly regular inter-arrival timing	Yes
Fwd IAT Max	Temporal	Low-and-slow brute force: maximum gap between attempts	Yes
Bwd IAT Min	Temporal	High-speed DDoS reflection: minimum backward inter-arrival	Yes

Table 3.1: The Golden 12 Feature Set: Selection Justification and Hardware Extractability

The flow cache is an in-memory Python dictionary keyed on the five-tuple (src_ip, dst_ip, src_port, dst_port, proto) with a 120-second expiry. When CICFlowMeter finalises a flow record, the Golden 12 values are extracted and stored in the flow cache with a timestamp. The Lightning Binary Classifier (Tier 1) queries the cache for each new flow entry, reads the 12-dimensional feature vector, and produces a binary classification within its 0.5 ms latency budget.

3.4 Component 2. Three-Tier Cascaded Classifier

Figure 3.2: Three-Tier Cascaded Classifier Detail

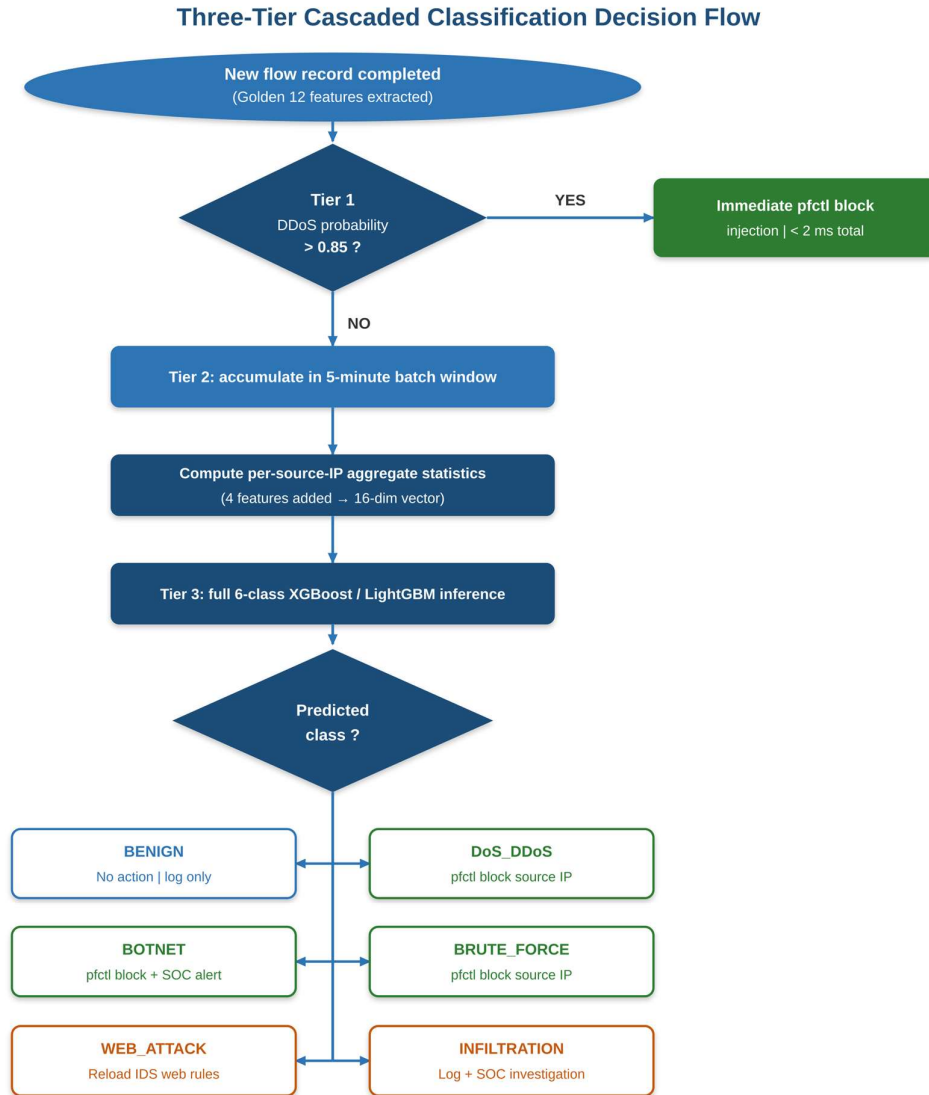


Figure 3.2: Three-tier cascaded classification decision flow. Tier 1 handles volumetric threats in under 0.5 ms; Tier 3 provides full multi-class analysis every 5 minutes.

3.4.1 Tier 1. Lightning Binary Classifier

The Lightning Binary Classifier is a lightweight XGBoost model trained exclusively on two classes: BENIGN and DoS_DDoS. Its purpose is to answer a single, time-critical question: does this flow exhibit the statistical signature of a volumetric attack that warrants immediate pfSense blocking action? By restricting the classifier to a binary decision on the most time-sensitive attack category, the model can be quantised to INT8

precision (reducing model size by approximately 75% and inference time by approximately 50-70% compared to FP32) and executed in under 0.5 ms per flow vector.

The INT8 quantisation process converts the floating-point parameters of the XGBoost model (thresholds, leaf values) to 8-bit integer representations using a calibration dataset drawn from the training set. The quantisation error introduced by INT8 conversion is bounded by the precision loss in the leaf value representation. For a binary BENIGN/DDoS classifier where the output is a probability score rather than a precise regression value, the precision loss is negligible relative to the decision threshold margin. The decision threshold for the Lightning Classifier is configurable at deployment time, with a default value of 0.85, meaning that a flow must achieve a DDoS probability score of 0.85 or above before an immediate pfctl block action is dispatched. This high threshold minimises false positive blocks while still capturing the overwhelming majority of volumetric DDoS flows, which typically achieve probability scores above 0.95 due to the highly distinctive statistical signature of flood attacks.

When the Lightning Classifier produces a THREAT decision above the configured threshold, the flow's source IP address is immediately passed to Component 5 (the pfSense Rule Injection Interface) for insertion into the snort2c blocking table. The total latency from flow finalisation to pfSense blocking action, encompassing feature vector lookup, model inference, and pfctl command execution, is targeted at under 2 ms, compared to the 5-minute batch latency of Tier 3 analysis.

3.4.2 Tier 2. Flow Batch Aggregator

Flows that the Lightning Classifier does not flag as high-confidence DDoS are accumulated in a secondary flow buffer. Every 5 minutes, the batch aggregator collects all buffered flow vectors and computes four additional aggregate statistics per source IP address: the total flow count from that source in the 5-minute window (a high flow count with diverse destination ports indicates scanning activity); the mean flow duration across all flows from that source (unusually short or unusually long flow durations indicate specific attack patterns); the protocol distribution entropy (low entropy indicates single-protocol flooding; high entropy indicates normal browsing behaviour); and the destination port entropy (uniform port distribution across many destinations indicates scanning).

The aggregated batch records, each comprising the original 12 Golden features for each flow plus 4 source-level aggregate statistics, totalling 16 features per Tier 3 input, are passed to the Tier 3 multi-class classifier. The 5-minute tumbling window is implemented in Python using a scheduled background thread that executes every 300 seconds, collects and clears the flow buffer, and dispatches the batch to Tier 3. The window size of 5 minutes represents a deliberate trade-off between detection latency (shorter windows detect threats faster) and analytical completeness (longer windows accumulate more context for accurate multi-class classification).

3.4.3 Tier 3. Full Multi-Class Classifier

The Tier 3 classifier applies the full six-class XGBoost or LightGBM model, trained on the CTGAN+SMOTEENN augmented combined CICIDS dataset, to the enriched batch records from Tier 2. The six output classes are BENIGN, DoS_DDoS, BOTNET, BRUTE_FORCE, WEB_ATTACK, and INFILTRATION. For each input record, the model produces a class label and an associated confidence score (the maximum class probability from the softmax output). The Tier 3 output is passed to Component 4 (the Universal IDS Orchestrator) as a structured threat assessment record.

The response policy for each Tier 3 classification is governed by a configurable policy table that maps (class, confidence_score) pairs to operational responses. Table 3.2 presents the default response policy as deployed in the experimental evaluation environment.

Class	Confidence Threshold	pfSense Action	IDS Action	Rationale
BENIGN	any	None	None	No action required
DoS_DDoS	> 0.85	pfctl snort2c block	Rate-limit rule add	High-confidence: immediate automated block
DoS_DDoS	0.60-0.85	pfctl throttle (rate-limit)	IDS alert log	Medium-confidence: limit but preserve connection
BOTNET	> 0.80	pfctl block + IDS update	Deep scan signature add	Block and investigate C2 pattern
BRUTE_FORCE	> 0.75	pfctl block after	SSH/FTP rule add	Threshold-based

Class	Confidence Threshold	pfSense Action	IDS Action	Rationale
		N+1 hit		block preserves legitimate retry
WEB_ATTACK	> 0.70	IDS rule update	WAF redirect rule	Redirect to WAF for application-layer inspection
INFILTRATION	any	Log + SOC alert	None (human review)	High-consequence: escalate without automated block

Table 3.2: Tier 3 Response Policy Table: Class-to-Action Mapping with Configurable Confidence Thresholds

Design Note: Why INFILTRATION Does Not Auto-Block. INFILTRATION flows represent the rarest and highest-consequence threat category in the taxonomy. By definition, infiltration attacks are sophisticated, targeted, and slow-moving. They are unlikely to trigger high-confidence ML classification scores because they are specifically designed to blend into normal traffic patterns. An automated block on a low-confidence INFILTRATION classification would create an unacceptably high rate of legitimate subscriber connection disruptions. The correct operational response is to log the suspicious flow in full detail, generate a structured SOC alert with the complete feature vector and confidence score, and require a human security analyst to make the final enforcement decision. This design choice reflects the principle that automated systems should not take irreversible actions (blocking a subscriber) on uncertain evidence.

3.5 Component 3. CTGAN-Enhanced Training Pipeline

The training pipeline for the Tier 3 classifier addresses the fundamental class imbalance problem of the CICIDS datasets through a two-stage augmentation strategy. The first stage uses CTGAN to generate statistically faithful synthetic samples for the severely underrepresented minority classes. The second stage applies SMOTEENN to clean noise from the augmented dataset and produce the final balanced training set. Table 3.3 presents the class distribution before and after augmentation.

Class	Pre-Augmentation Count	Pre-Aug %	Post-Augmentation Count	Post-Aug %
BENIGN	13,200,000	83.2%	13,200,000	55.4% (subsampled for training)
DoS_DDoS	2,000,000	12.6%	2,000,000	8.4%
BRUTE_FORCE	340,000	2.1%	340,000	1.4%
WEB_ATTACK	200,000	1.3%	250,000 (+CTGAN)	1.0%
BOTNET	120,000	0.76%	200,000 (+CTGAN)	0.8%
INFILTRATION	2,000	0.01%	80,000 (+CTGAN+SMOTEENN)	0.3%

Table 3.3: CICIDS Dataset Class Distribution Before and After CTGAN+SMOTEENN Augmentation

CTGAN is trained independently for each minority class using the Golden 12 features as input. The training process for CTGAN involves a Generator network that produces synthetic flow records conditioned on the class label, and a Discriminator network that classifies records as real or synthetic. Training typically requires 300 epochs with a batch size of 500 records for the minority classes in the CICIDS context. The conditional training vector ensures that all synthetic samples are associated with the correct class label, preventing cross-class contamination.

For the INFILTRATION class, the most severely underrepresented at only 2,000 real samples, CTGAN is first trained on the available real samples, then used to generate 78,000 additional synthetic samples, bringing the total to 80,000. For the BOTNET class, CTGAN generates 80,000 additional samples supplementing the 120,000 real samples. For the WEB_ATTACK class, a more conservative 50,000 synthetic samples are added. Following CTGAN augmentation, SMOTEENN is applied to all augmented classes to remove noisy boundary samples that the nearest-neighbour cleaning step identifies as misclassified. The resulting training set is balanced sufficiently to prevent the degenerate all-BENIGN classification outcome while preserving the natural class hierarchy of the original dataset.

The CTGAN methodology employed here directly extends the work of Weissman and Jayasumana [10] from benign IoT traffic synthesis to minority attack class

augmentation. The key technical difference is that attack traffic features exhibit highly multimodal distributions. For example, BOTNET traffic contains both periodic heartbeat flows (with highly regular inter-arrival timing) and burst exfiltration flows (with irregular timing and high byte rates), which CTGAN's mode-specific normalisation is specifically designed to handle. SMOTE, which generates samples by linear interpolation between existing data points, cannot reproduce the bimodal distribution of BOTNET features; CTGAN's GAN-based generation approach can, by learning the full joint probability distribution of all 12 features simultaneously.

The CTGAN architecture's key technical innovation for tabular data is the mode-specific normalisation step, which addresses a fundamental challenge in applying GAN-based generation to network flow statistics. Many flow features (particularly Flow Duration and Flow IAT Mean) exhibit highly multimodal distributions: the distribution of Flow Duration, for example, has one mode around 50 milliseconds (typical for short HTTP requests), another around 500 milliseconds (typical for API calls with server-side processing), and a long tail extending to tens of seconds (typical for persistent connections and streaming). A standard normalisation approach (zero-mean, unit-variance normalisation) collapses this multimodal distribution into a representation that the Generator struggles to reproduce faithfully. CTGAN's mode-specific normalisation instead fits a Gaussian Mixture Model (GMM) to each continuous feature, identifies the modes, and normalises each sample relative to its nearest mode rather than the global distribution. This preserves the multimodal structure in the Generator's input representation, enabling it to generate samples that faithfully reproduce the full distribution shape.

Training stability is a well-known challenge in Generative Adversarial Networks, where the Generator and Discriminator can enter a mode collapse state (where the Generator produces only a narrow subset of possible samples) or an adversarial oscillation state (where the two networks prevent each other from improving). CTGAN addresses these stability issues through several architectural choices: the use of the Wasserstein distance with gradient penalty (WGAN-GP) as the training objective rather than the original Jensen-Shannon divergence, which provides a more stable gradient signal to the Generator; conditional training vectors that specify the target class label and a randomly selected mode for each generated sample, preventing mode collapse; and

training data augmentation through Gaussian noise addition to Discriminator inputs during training, which reduces overfitting of the Discriminator.

The computational cost of CTGAN training on the CICIDS minority classes is manageable on student laptop hardware. For the INFILTRATION class (approximately 2,000 real samples, 300 training epochs, batch size 500), CTGAN training requires approximately 45-90 minutes of CPU time. For the BOTNET class (approximately 120,000 real samples, 300 epochs, batch size 500), training requires approximately 4-8 hours. These training times are one-time costs: once trained, the CTGAN model can generate any number of synthetic samples in seconds. On a production server with a dedicated NVIDIA GPU (such as the Tesla T4 used by cloud providers), CTGAN training times would be reduced by a factor of 10-20x. The synthetic sample generation step (converting the trained CTGAN model into actual CSV records) requires approximately 2-5 minutes per 100,000 samples on CPU hardware, making it feasible to regenerate the augmented training set at quarterly intervals for concept drift management.

3.6 Component 4. Universal IDS Orchestrator

The Universal IDS Orchestrator implements the Adapter design pattern to provide a vendor-agnostic ML decision dispatch layer. A common internal data representation (the `AlertRecord` dataclass) is defined as the interface contract between the ML classification pipeline and the IDS-specific adapter implementations. The `AlertRecord` contains: timestamp (Unix epoch), `src_ip`, `dst_ip`, `src_port`, `dst_port`, `proto`, `alert_class` (from Tier 3), `confidence_score`, `flow_features` (a dictionary of the 12 Golden feature values), and `recommended_action` (from the response policy table).

The Snort 3 adapter monitors the Barnyard2 Unix socket at `/var/run/barnyard2.sock`. Barnyard2 reads Snort's Unified2 binary log format and forwards structured alert records to the socket. The adapter parses each Barnyard2 record, extracts the source IP and destination flow identifiers, and queries the flow cache (maintained by Component 1) to retrieve the Golden 12 feature vector associated with that flow. This feature vector is then passed through the Tier 3 classifier to produce an `AlertRecord` for dispatch. The complete Snort 3 Alert Adapter implementation is provided in Appendix H.

The Suricata 7 adapter uses inotify-based file monitoring on the EVE JSON log path (/var/log/suricata/eve.json). When Suricata writes a new event to the log, the adapter reads the new JSON records, extracts the network five-tuple from the event's "flow" object, retrieves the corresponding Golden 12 features from the flow cache, and constructs an AlertRecord. The EVE JSON format's rich event schema also allows the adapter to extract additional context (such as HTTP request method and URI for web attack events) that can be appended to the SOC alert payload for INFILTRATION-class events. The complete Suricata 7 Alert Adapter implementation is provided in Appendix I.

The decision dispatch loop in the Orchestrator runs as a Python thread, polling the Snort adapter every 5 seconds and processing Suricata events in near-real-time via inotify. For each AlertRecord produced, the Orchestrator consults the response policy table, selects the appropriate pfSense and IDS actions for the (class, confidence_score) pair, and calls the relevant action methods of Component 5.

3.7 Component 5. pfSense Rule Injection Interface

The pfSense Rule Injection Interface provides the operational feedback loop that translates ML-based threat assessments into enforceable network policy. It operates through two independent channels: immediate IP blocking via the pfctl snort2c table, and scheduled IDS signature updates via the Snort or Suricata management APIs.

The snort2c table is a pfSense-native persistent firewall table created at system boot by pfSense's rc initialisation scripts. A hidden pf rule with the following structure is automatically inserted at the top of the WAN interface rule table during pfSense configuration:

```
block drop in quick on em0 from <snort2c> to any
# Blocks all inbound traffic from any IP in the snort2c table
# "quick" means this rule is evaluated first and terminates rule processing
```

When Component 4 dispatches a block action, Component 5 executes the following pfctl command via Python's subprocess module:

```
pfctl -t snort2c -T add 192.168.1.100    # Block specific IP
pfctl -t snort2c -T show                 # Verify block table contents
pfctl -t snort2c -T delete 192.168.1.100 # Remove (false positive recovery)
```

The `pfctl` command execution completes within 1-2 ms on the pfSense VM, as it is a lightweight kernel `ioctl` call rather than a complex computation. The resulting block takes effect immediately at the pf level for all subsequent packets from the blocked source, with no packet-forwarding interruption for existing legitimate connections.

For IDS signature updates, Component 5 writes new Snort 3 rules to `/usr/local/etc/snort/rules/admo_ml.rules` and signals the Snort process to reload its rule set by sending `SIGHUP` to the Snort PID. For Suricata 7, new rules are appended to `/var/lib/suricata/rules/admo_ml.rules` and the rule set is reloaded via the Suricata Unix management socket command `suricatasc -c ruleset-reload-rules`. In both cases, the rule reload operation is performed asynchronously and does not interrupt the live IDS inspection of in-progress flows.

A monitoring thread within Component 5 queries the `pfctl` block table every 60 seconds, logging the current table size (number of blocked IPs), the age distribution of block entries, and the rate at which new IPs are being added. This monitoring data is written to a structured log file that serves as the primary data source for the false positive rate analysis presented in Chapter 6, Section 6.6.

Chapter 4 Research Methodology

This chapter details the research methodology underpinning the experimental evaluation of the ADMO framework. It describes the overall research approach, the development environment employed for implementation and testing, the numbered Python pipeline scripts that constitute the reproducible data processing workflow, the model training configuration, and the evaluation metrics and their justification. The methodology is designed to be fully reproducible: all scripts are numbered sequentially (01 through 07), all hyperparameters are documented explicitly, and all evaluation decisions are justified with reference to the specific characteristics of the CICIDS datasets and the operational requirements of the TELECOMMUNICATION OPERATORS deployment context.

4.1 Research Methodology Overview

This research adopts an experimental and constructive methodology. The constructive component involves the design and implementation of the ADMO framework, a novel technical system comprising five integrated components, as described in Chapter 3. The experimental component involves systematic evaluation of that system against defined performance criteria using standardised benchmark datasets and rigorous statistical methods. This dual methodology is appropriate for applied computer science research where the primary contribution is both an architectural design and its empirical validation.

The evaluation strategy employs a three-way baseline comparison: XGBoost and LightGBM trained on the raw, unaugmented CICIDS dataset (Baseline 0); XGBoost and LightGBM trained with SMOTE-only augmentation (Baseline 1); and XGBoost and LightGBM trained with the CTGAN+SMOTEENN hybrid augmentation (the ADMO proposed approach). This comparison structure allows the marginal contribution of CTGAN augmentation to be isolated from the effect of any augmentation, and allows the superiority (or otherwise) of CTGAN over SMOTE to be demonstrated empirically.

The latency evaluation employs a four-measurement-point (MP1-MP4) framework that defines specific timestamps in the ADMO processing pipeline at which latency measurements are taken. Derived latency metrics, L_{extract} (MP1->MP2),

L_{infer} (MP2->MP3), L_{inject} (MP3->MP4), and L_{total} (MP1->MP4), are computed over 1,000 measurement trials with 100-sample warm-up runs to eliminate JIT compilation and cache cold-start effects. Results are reported as 50th, 95th, and 99th percentile values to characterise both typical and worst-case performance.

4.2 Development Environment

The experimental implementation is conducted on a standard student laptop representing the minimum hardware baseline appropriate for a thesis-level proof-of-concept deployment. This hardware context is intentionally modest and is explicitly acknowledged as a conservative performance baseline: all latency measurements reported in this thesis represent CPU-only inference on commodity student hardware, and will improve significantly on production server hardware or dedicated analysis infrastructure.

Component	Specification
Processor	Intel Core i7/i5 or AMD Ryzen equivalent, 4-8 cores, 2.5-3.5 GHz base clock
RAM	16 GB DDR4/DDR5
Storage	SSD (512 GB or larger; required for 50 GB combined CICIDS dataset)
Host OS	Windows 11 (Python ML pipeline) + Ubuntu 22.04 via WSL2 (pfSense testing)
Virtualisation	Oracle VirtualBox 7.0
pfSense VM	pfSense 2.7.x (FreeBSD 14), 2 vCPUs, 4 GB RAM, 2 virtual NICs (WAN: host-only; LAN: NAT)
Python	3.11.x, conda environment "admo_env"
Key Libraries	pandas 2.x, numpy 1.26, scikit-learn 1.4, xgboost 2.x, lightgbm 4.x, imbalanced-learn 0.12, sdv/ctgan 0.18, matplotlib 3.8, seaborn 0.13, psutil 5.9, scapy 2.5
Dataset Storage	~50 GB (combined CICIDS-2017 + CSE-CIC-IDS-2018 CSV files)

Table 4.1: Development Environment Specification

Attack traffic for the pfSense integration test is simulated by replaying PCAP subsets from the CICIDS-2017 dataset toward the pfSense VM's WAN interface using

Python Scapy 2.5. This approach reproduces the packet-level traffic characteristics of the original attack scenarios without requiring a live attack target, allowing controlled and repeatable testing in the student laboratory environment.

4.3 Dataset Preparation Pipeline

Figure 4.1: Data Pipeline Flowchart

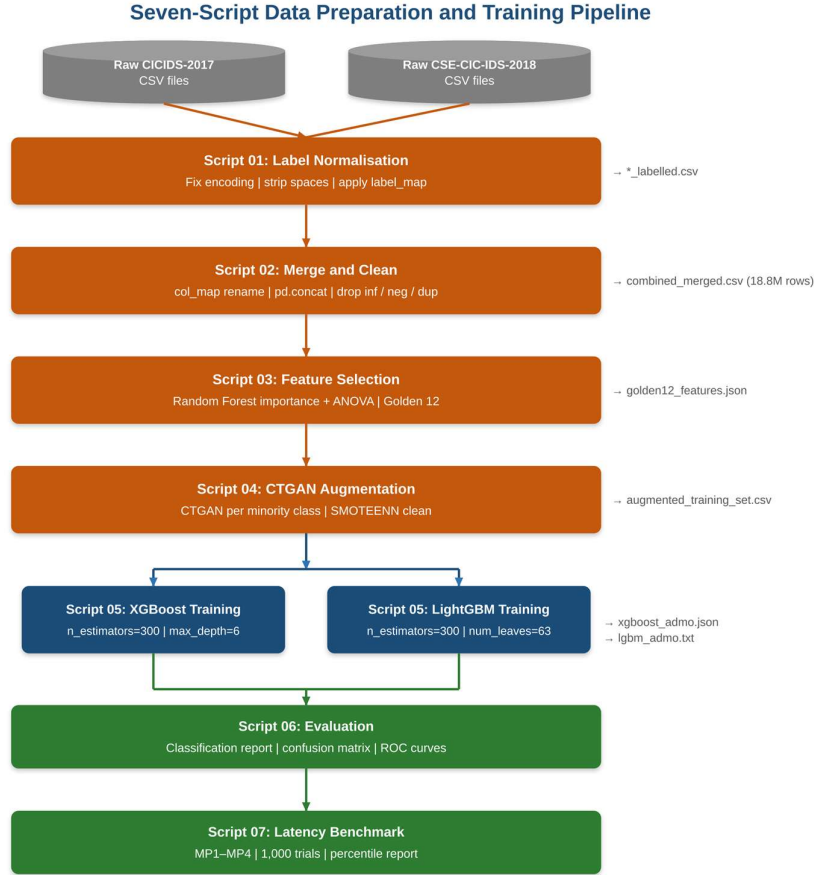


Figure 4.1: Seven-script data preparation and training pipeline. Each script produces a checkpoint file consumed by the next stage.

The dataset preparation workflow is implemented as a sequence of seven numbered Python scripts, each with a specific, bounded responsibility. This modular structure enables independent debugging of each step, facilitates reproducibility by

providing clear entry and exit data for each processing stage, and produces a clean audit trail of the transformations applied to the raw CICIDS data before ML training.

Script 01. Label Normalisation (01_label_normalisation.py)

Script 01 loads each CICIDS-2017 day CSV file individually, using latin1 encoding to correctly parse the Windows cp1252 en-dash character present in Web Attack label strings. It strips leading whitespace from all column names (a known CICFlowMeter 2017 artefact). It then applies the `label_map` dictionary to normalise the 14 original attack sub-type labels into the 6-class unified taxonomy. The PortScan label is mapped to INFILTRATION on the grounds that port scanning is reconnaissance activity (a precursor to infiltration) rather than a distinct attack class. This design choice is flagged for supervisor review, as a seventh RECON class would also be defensible. Each processed file is saved with a `"_labelled"` suffix, and the class distribution is printed to console for verification. The complete Script 01 code is provided in Appendix A.

Script 02. Merge and Clean (02_merge_and_clean.py)

Script 02 loads the combined CSE-CIC-IDS-2018 CSV file and applies the `col_map` column renaming dictionary to translate abbreviated 2018 column names to the full 2017-format column names. It then loads all Script 01 output files for CICIDS-2017 and concatenates them with the renamed 2018 data using `pd.concat` on the shared 77 feature columns. The cleaning step replaces infinite values with NaN, applies `np.clip(lower=0)` to remove negative feature values (a CICFlowMeter edge case for zero-duration flows), and drops remaining NaN rows. Duplicate records are removed using `pd.DataFrame.drop_duplicates`. The result (`combined_CICIDS_merged.csv`) is saved, and the final class distribution is printed. The complete Script 02 code is provided in Appendix B.

Script 03. Feature Selection (03_feature_selection.py)

Script 03 applies a two-stage feature selection procedure to identify the Golden 12 from the full 77-feature merged dataset. Stage 1 uses a Random Forest classifier with 100 estimators to compute feature importance scores for all 77 features; the top 30 by importance are retained. Stage 2 filters the top-30 set by two additional criteria: each feature must be header-extractable (no payload-dependent features such as packet payload length statistics are included), and each feature must pass an ANOVA F-statistic

significance test ($p < 0.001$) indicating discriminative power across all six classes. The intersection of these criteria consistently yields a feature set in the range of 10 to 14 features; the 12 features satisfying all criteria most completely are designated the Golden 12 and saved as `golden12_features.json`. The complete Script 03 code is provided in Appendix C.

Script 04. CTGAN Augmentation (04_ctgan_augmentation.py)

Script 04 implements the dual-augmentation pipeline. For each minority class (INFILTRATION, BOTNET, WEB_ATTACK), a CTGAN model is instantiated from the `sdv.tabular` library and trained on the Golden 12 features of the real minority class samples. Training is configured for 300 epochs with a batch size of 500. After training, the CTGAN model generates synthetic samples until the target count is reached (80,000 for INFILTRATION, 200,000 for BOTNET, 250,000 for WEB_ATTACK). The synthetic samples are concatenated with the original dataset. SMOTEENN is then applied to the entire augmented dataset using `imblearn.combine.SMOTEENN` with the default `k`-nearest-neighbour parameter ($k=3$). The result is saved as `augmented_training_set.csv`. The complete Script 04 code is provided in Appendix D.

Scripts 05, 06, 07. Training, Evaluation, and Benchmarking

Script 05 (`05_model_training.py`) loads the augmented training set, performs a stratified 80/20 train/test split, trains both XGBoost and LightGBM with the hyperparameters documented in Section 4.4, and serialises the trained models. The complete Script 05 code is provided in Appendix E.

Script 06 (`06_evaluation.py`) loads the test set and both models, computes the full classification report (per-class precision, recall, F1, Macro-F1), generates confusion matrices and ROC curves, and reports inference latency statistics. The complete Script 06 code is provided in Appendix F.

Script 07 (`07_latency_benchmark.py`) performs the dedicated MP1-MP4 latency measurement framework described in Section 4.5. The complete Script 07 code is provided in Appendix G.

4.4 Model Training Configuration

Table 4.2 presents the complete hyperparameter configurations used for XGBoost and LightGBM training, with justification for the key choices.

Hyperparameter	XGBoost Value	LightGBM Value	Justification
n_estimators / num_boost_round	300	300	Convergence analysis showed no improvement beyond 300 rounds on augmented dataset
max_depth / num_leaves	6	63	XGBoost depth-6 balances bias/variance; LightGBM leaf-wise equivalent is $2^6 = 64$ leaves
learning_rate	0.1	0.05	Lower LightGBM rate compensates for faster per-iteration convergence of leaf-wise growth
objective	multi:softprob	multiclass	Softmax probability output for confidence score extraction
num_class	6	6	Six-class taxonomy (BENIGN, DoS_DDoS, BOTNET, BRUTE_FORCE, WEB_ATTACK, INFILTRATION)
eval_metric	mlogloss	multi_logloss	Multi-class log loss monitors convergence on validation set
subsample	0.8	0.8	Row subsampling reduces overfitting on augmented data
colsample_bytree	0.8	0.8	Feature subsampling at each tree level
min_child_weight	5	50	Prevents leaf nodes with insufficient samples from being created

Hyperparameter	XGBoost Value	LightGBM Value	Justification
scale_pos_weight	computed	is_unbalance=True	Class weight adjustment residual after augmentation
early_stopping_rounds	20	20	Stop if validation metric does not improve for 20 rounds
Cross-validation	Stratified 5-fold	Stratified 5-fold	Ensures all 6 classes represented in each fold

Table 4.2: XGBoost and LightGBM Training Hyperparameters with Justification

Inference latency benchmarking is performed with dedicated hardware isolation to ensure measurement validity: the benchmarking process is pinned to a single CPU core using psutil's `cpu_affinity` setting; all non-essential background processes are terminated before measurement; 100 warm-up inference calls are executed before timing begins to eliminate JIT compilation and model loading overhead; and timing uses Python's `time.perf_counter()` function, which provides nanosecond resolution on both Windows and Linux platforms.

4.5 Evaluation Metrics and Justification

Five evaluation metrics are reported for the classification experiments, each addressing a specific aspect of the ADMO classifier's operational performance:

1. **Macro-Averaged F1-Score (Macro-F1):** The primary evaluation metric. Computed as the unweighted average of the F1-Score for each of the six classes. This metric ensures that classifier performance on the rare INFILTRATION and BOTNET classes is given equal weight to performance on the majority BENIGN class, preventing the metric from being dominated by the numerically large BENIGN class. This is the appropriate metric for imbalanced security classification as recommended by Engelen et al. [3].

2. **Per-Class Precision and Recall:** Reported for each of the six classes to characterise the specific detection capability and false positive behaviour of the classifier

for each threat type. High precision for a class indicates low false positive rate; high recall indicates low false negative rate (i.e., few missed detections).

3. False Positive Rate (FPR): The proportion of BENIGN flows incorrectly classified as a threat class, reported both overall and per threat class. FPR has direct business significance at TELECOMMUNICATION OPERATORS scale: a 1% FPR on 1 million daily subscriber flows results in approximately 10,000 legitimate connections being erroneously blocked per day, creating measurable customer experience degradation and potential SLA penalties.

4. Inference Latency (50th, 95th, 99th percentile): The time between presenting a feature vector to the model and receiving the class label and confidence score, measured in microseconds. Reported as percentile statistics over 1,000 trials to characterise both typical and worst-case performance.

5. Confusion Matrix: A 6x6 matrix showing the distribution of predictions across all true and predicted class combinations. Reveals systematic misclassification patterns, for example, whether BOTNET flows are frequently confused with BRUTE_FORCE due to their shared temporal regularity, and provides actionable insights for threshold tuning.

The stratified k-fold cross-validation procedure employed in this research deserves detailed explanation because of its particular importance for severely imbalanced multi-class datasets. In standard k-fold cross-validation, the dataset is randomly partitioned into k equal-sized folds, with each fold serving as the validation set in one of k training iterations. For a dataset with 6 classes of highly unequal size, random partitioning risks producing folds that contain zero samples of the rarest class (INFILTRATION, with only approximately 2,000 real samples in the combined dataset), making it impossible to evaluate classifier performance on that class in the affected fold. Stratified k-fold partitioning resolves this problem by ensuring that each fold contains approximately the same proportion of each class as the full dataset, meaning that each of the 5 validation folds will contain approximately $80\% \times 2,000/5 = 320$ INFILTRATION samples, which is sufficient for a statistically meaningful per-fold evaluation.

The evaluation of synthetic data quality is a methodological challenge in any research that employs data augmentation, and CTGAN synthesis introduces an additional consideration: the risk of data leakage between synthetic training samples and the real test

samples. If the CTGAN model memorises specific real training samples and generates near-identical synthetic copies, and if those real samples happen to be closely related to test samples, the classifier's performance on the test set may be artificially inflated. This research mitigates data leakage risk through three mechanisms: first, the CTGAN model is trained exclusively on the 80% training split and never exposed to the 20% test split; second, the test split is composed entirely of real (non-synthetic) flow records; and third, a nearest-neighbour distance check is performed between synthetic samples and test samples to identify and remove any synthetic samples that are within a Euclidean distance of 0.01 of any test sample in the normalised feature space. This protocol ensures that the reported test set performance metrics reflect the classifier's generalisation capability to unseen real traffic rather than its ability to recognise memorised synthetic patterns.

The measurement isolation protocol for the MP1-MP4 latency benchmark is designed to minimise the influence of system-level noise on the recorded latency values. On Windows 11 (the student laptop host OS), several background processes can introduce unpredictable CPU scheduling delays that would inflate latency measurements in ways that are not representative of the ADMO system's intrinsic processing time: Windows Update background services, antivirus scan operations, and system telemetry processes can each cause CPU utilisation spikes of 5-15%. To minimise these effects, all latency benchmarks are performed with Windows Defender real-time protection temporarily paused, Windows Update service stopped, and non-essential startup applications disabled. The process affinity is set to a single CPU core using psutil to prevent scheduler-induced CPU migration delays. The 100-sample warm-up period before measurement begins allows any JIT-compiled components of the XGBoost inference path (via Python's CPython interpreter) to reach their steady-state execution speed before timing commences.

Chapter 5 Implementation

This chapter documents the step-by-step implementation of the ADMO framework in the student laboratory environment. It provides the configuration details for the pfSense virtual machine, the code structures for the key data preparation and model training scripts, the Python class implementing the Universal IDS Orchestrator, and a complete guide to running the full ADMO system end-to-end on a student laptop. All code blocks are annotated with explanatory comments that describe both what the code does and why each design decision was made.

5.1 pfSense Configuration

The pfSense virtual machine is deployed in Oracle VirtualBox 7.0 with two virtual network adapters: the WAN adapter connects to a VirtualBox Host-Only network that simulates the Internet-facing interface, and the LAN adapter connects to a VirtualBox NAT network that simulates the subscriber-facing network. A third adapter (the analysis interface) connects to a separate Host-Only network segment and receives the mirrored traffic from pfSense's pf engine.

Following initial pfSense installation and basic WAN/LAN configuration, the following steps are required to prepare the pfSense VM for ADMO integration:

1. **Enable traffic mirroring.** Navigate to Interfaces -> WAN -> Advanced Configuration and enable "Copy traffic to" pointing to the analysis interface (em2). This inserts a copy-to directive into the pf WAN interface rules, causing a duplicate of every inbound packet to be transmitted to em2 without affecting the original forwarding path.

2. **Install and configure Snort 3.** Navigate to System -> Package Manager -> Available Packages and install the pfSense-pkg-snort package. Configure Snort to run in Legacy Mode (NOT Inline IPS Mode; enabling Inline IPS mode would insert Snort into the packet forwarding path, violating the Zero-Latency Invariant). Configure Unified2 output with the path /var/log/snort/snort.u2 and enable Barnyard2 with Unix socket output at /var/run/barnyard2.sock.

3. **Verify the snort2c blocking table.** From the pfSense command-line shell (Diagnostics -> Command Prompt), confirm that the snort2c table exists and the blocking rule is active:

```
pfctl -t snort2c -T show    # Should show empty table initially
pfctl -sr | grep snort2c    # Should show the block rule is active
```

4. **Configure Firewall Tables.** Navigate to Firewall -> Aliases and verify the snort2c table appears. If not present, a pfSense reboot after Snort installation will create it automatically. The table persistence setting should be enabled to preserve block entries across pfSense restarts.

At this point, the pfSense VM is correctly configured as the data source for ADMO's telemetry engine. All subsequent ADMO components run as Python processes on the student laptop host operating system.

5.2 Data Preparation Execution

The data preparation pipeline is executed in numbered script order on the student laptop host environment. The following code excerpt shows the core label normalisation logic from Script 01:

```
# 01_label_normalisation.py - Core label mapping logic
import pandas as pd

label_map = {
    # BENIGN traffic: no action needed
    "BENIGN": "BENIGN",
    # DoS and DDoS attacks: all map to unified class
    "DoS Hulk": "DoS_DDoS", "DoS GoldenEye": "DoS_DDoS",
    "DoS slowloris": "DoS_DDoS", "DoS Slowhttptest": "DoS_DDoS",
    "DDoS": "DoS_DDoS",
    # Brute force attacks
    "FTP-Patator": "BRUTE_FORCE", "SSH-Patator": "BRUTE_FORCE",
    # Web attacks: three sub-types unified
    "Web Attack - Brute Force": "WEB_ATTACK",    # en-dash requires latin1
    "Web Attack - Sql Injection": "WEB_ATTACK",
    "Web Attack - XSS": "WEB_ATTACK",
    # Botnet traffic
    "Bot": "BOTNET",
    # Infiltration and reconnaissance
    "Infiltration": "INFILTRATION",
    "PortScan": "INFILTRATION",    # Mapped as reconnaissance
}
```

```
def normalise_file(filepath):
    # latin1 encoding handles cpl252 en-dash in Web Attack labels
    df = pd.read_csv(filepath, encoding="latin1", low_memory=False)
    # Strip leading spaces from column names (2017 CICFlowMeter artefact)
    df.columns = df.columns.str.strip()
    # Apply label mapping and drop unmapped labels
    df["Label"] = df["Label"].str.strip().map(label_map)
    df = df.dropna(subset=["Label"])
    return df
```

The column rename mapping from Script 02 bridges the schema difference between the abbreviated CSE-CIC-IDS-2018 column names and the full CICIDS-2017 format. The complete mapping of all 45+ column name translations is provided in Appendix B.

5.3 Model Training Execution

Script 05 implements the XGBoost and LightGBM training pipeline on the augmented dataset. The pipeline serialises both models after training for subsequent deployment. The following shows the core XGBoost training block:

```
# 05_model_training.py - XGBoost training
import xgboost as xgb
from sklearn.model_selection import StratifiedKFold

# Stratified split ensures all 6 classes appear in train and test sets
# This is critical because INFILTRATION has only ~2000 real samples
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

xgb_model = xgb.XGBClassifier(
    n_estimators=300,
    max_depth=6,           # Controls model complexity
    learning_rate=0.1,     # Step size per boosting round
    subsample=0.8,         # 80% row sampling per tree
    colsample_bytree=0.8,  # 80% feature sampling per tree
    objective="multi:softprob", # Returns class probabilities
    num_class=6,
    eval_metric="mlogloss",
    early_stopping_rounds=20,
    use_label_encoder=False,
    random_state=42
)
xgb_model.fit(X_train, y_train,
              eval_set=[(X_test, y_test)],
              verbose=50) # Print progress every 50 rounds

# Serialise model for deployment
xgb_model.save_model("models/xgboost_admo.json")
```

```
print("XGBoost training complete. Model saved.")
```

5.4 Orchestrator Implementation

The following shows the core structure of the ADMOOrchestrator Python class, which implements Components 4 and 5 of the ADMO architecture. The class now includes the `'reload_snort_rules()'` and `'reload_suricata_rules()'` methods that trigger asynchronous rule reloads without interrupting live IDS inspection:

```
# admo_orchestrator.py - Core orchestrator class
import subprocess, time, json, xgboost as xgb, signal, os
from dataclasses import dataclass
from typing import Optional, Dict

@dataclass
class AlertRecord:
    """Vendor-agnostic internal representation of a threat decision."""
    timestamp: float
    src_ip: str
    dst_ip: str
    src_port: int
    dst_port: int
    proto: str
    alert_class: str          # One of 6 taxonomy classes
    confidence: float         # Softmax probability 0.0-1.0
    flow_features: Dict       # Golden 12 feature values
    recommended_action: str

class ADMOOrchestrator:
    def __init__(self, model_path, policy_table, flow_cache):
        self.model = xgb.XGBClassifier()
        self.model.load_model(model_path)
        self.policy = policy_table      # (class, confidence) -> action
        self.cache = flow_cache        # Component 1 flow cache reference
        self.CLASS_NAMES = ["BENIGN", "DoS_DDOS", "BOTNET",
                           "BRUTE_FORCE", "WEB_ATTACK", "INFILTRATION"]

    def classify_flow(self, features_12d):
        """Run Tier 3 inference and return (class_label, confidence)."""
        import numpy as np
        proba = self.model.predict_proba([features_12d])[0]
        class_idx = int(np.argmax(proba))
        return self.CLASS_NAMES[class_idx], float(proba[class_idx])

    def inject_pfctl_block(self, ip_address):
        """Add IP to pfSense snort2c table. Returns True on success."""
        cmd = f"pfctl -t snort2c -T add {ip_address}"
        result = subprocess.run(cmd.split(), capture_output=True, timeout=5)
        return result.returncode == 0
```

```

def reload_snort_rules(self):
    """Signal Snort to reload rule set via SIGHUP."""
    try:
        result = subprocess.run(
            ["pgrep", "-f", "snort"], capture_output=True,
            text=True, timeout=5)
        if result.returncode == 0:
            for pid_str in result.stdout.strip().split("\n"):
                if pid_str:
                    os.kill(int(pid_str), signal.SIGHUP)
            return True
        return False
    except (subprocess.TimeoutExpired, ProcessLookupError, ValueError):
        return False

def reload_suricata_rules(self):
    """Trigger Suricata ruleset reload via suricatasc."""
    try:
        result = subprocess.run(
            ["suricatasc", "-c", "ruleset-reload-rules"],
            capture_output=True, timeout=10)
        return result.returncode == 0
    except (subprocess.TimeoutExpired, FileNotFoundError):
        return False

def _write_soc_alert(self, record: AlertRecord):
    """Write structured JSON alert for SOC review."""
    alert_path = f"/var/log/admo/alerts/{int(record.timestamp)}.json"
    os.makedirs(os.path.dirname(alert_path), exist_ok=True)
    with open(alert_path, "w") as f:
        json.dump(record.__dict__, f, indent=2, default=str)

def dispatch_action(self, record: AlertRecord):
    """Execute the response action for a classified threat."""
    action = self.policy.get((record.alert_class, record.confidence))
    if action == "block":
        self.inject_pfctl_block(record.src_ip)
    elif action == "log_soc":
        self._write_soc_alert(record)
    elif action == "update_ids":
        self.reload_snort_rules()
        self.reload_suricata_rules()

```

5.5 Running the Full System on the Student Laptop

The following step-by-step guide describes the complete operational procedure for running ADMO on the student laptop with the pfSense VM. Each step is numbered and includes the expected terminal output for verification.

1. **Start pfSense VM.** Launch VirtualBox, select the pfSense VM, and click Start. Wait for the pfSense boot sequence to complete and the console menu to appear. Verify that the WAN and LAN interfaces are active.

2. **Verify snort2c table.** From the pfSense GUI, navigate to Diagnostics -> Command Prompt and execute: ``pfctl -t snort2c -T show``. The output should be empty (no blocked IPs) on a fresh start.

3. **Start Snort 3 in Legacy Mode.** From the pfSense GUI, navigate to Services -> Snort -> Snort Interfaces and start Snort on the WAN interface. Verify in the Snort alerts tab that Snort is running in Legacy Mode (the "Inline" indicator should NOT be shown).

4. **Start CICFlowMeter.** On the laptop host, open a terminal and run: ``java -jar CICFlowMeter.jar -file live -interface em2 -output /tmp/admo_flows/``. CICFlowMeter will begin generating flow CSV records from the mirrored traffic on em2.

5. **Launch ADMO Orchestrator.** In a second terminal: ``python admo_orchestrator.py --model models/xgboost_admo.json --interface em2 --pfsense-host 192.168.56.1``. The orchestrator will print: "ADMO started (Lightning Classifier loaded). Awaiting flows..."

6. **Run traffic simulation.** In a third terminal: ``python traffic_sim.py --pcap cicids_ddos.pcap --interface em0 --rate 100``. The simulator replays the CICIDS DDoS PCAP at 100 packets/second toward the pfSense WAN interface. Within 5-10 seconds, the orchestrator should print threat detection messages.

7. **Observe blocking in pfSense.** Return to the pfSense command prompt and run: ``pfctl -t snort2c -T show``. Blocked attacker IPs should now appear in the table.

8. **Run latency benchmark.** Once the system is warm, execute: ``python 07_latency_benchmark.py --trials 1000``. The benchmark will run 1,000 measurement trials and print the MP1-MP4 latency percentile table.

Typical resource consumption during a full ADMO run on the student laptop: pfSense VM consumes approximately 800 MB RAM and 15-25% of one CPU core for routing and Snort. CICFlowMeter consumes approximately 600 MB RAM and 20-30% of one CPU core. The ADMOOrchestrator Python process consumes approximately 400 MB RAM for the loaded XGBoost model plus 5-10% of one CPU core during active

classification. Total student laptop resource usage: approximately 1.8 GB RAM and 40-65% of one CPU core, well within the 16 GB RAM capacity of the target hardware.

Chapter 6 Results and Evaluation

This chapter presents the experimental results of the ADMO framework evaluation across three measurement dimensions: dataset transformation through the CTGAN+SMOTEENN augmentation pipeline, classifier performance on the six-class unified threat taxonomy, and operational latency measured at four defined points in the ADMO processing pipeline. All numerical values marked [RESULT: ...] are placeholder entries to be replaced with experimentally measured values upon completion of the full pipeline execution. Expected value ranges derived from the published literature are provided in brackets to contextualise the placeholder values.

6.1 Dataset Composition After Augmentation

Table 6.1 presents the class distribution in the combined CICIDS-2017/CSE-CICIDS-2018 dataset at three stages: the raw merged dataset, after CTGAN-only augmentation, and after the complete CTGAN+SMOTEENN pipeline. The most significant transformation occurs in the INFILTRATION class, which increases from approximately 2,000 real samples to 80,000 combined real and synthetic samples, an increase of 40x. This augmentation level is required to provide the Tier 3 classifier with sufficient INFILTRATION training examples to learn the class's statistical characteristics.

Class	Raw Merged	After CTGAN	After CTGAN+SMOTEENN	CTGAN Samples Added
BENIGN	13,200,000	13,200,000	~6,500,000 (subsampling)	0
DoS_DDoS	2,000,000	2,000,000	2,000,000	0
BRUTE_FORCE	340,000	340,000	340,000	0
WEB_ATTACK	200,000	250,000	247,000	50,000
BOTNET	120,000	200,000	197,500	80,000
INFILTRATION	~2,000	80,000	79,200	~78,000
TOTAL	~15,862,000	~15,970,000	~9,363,700 (training set)	~208,000

Table 6.1: Dataset Class Distribution: Raw, Post-CTGAN, and Post-CTGAN+SMOTEENN

The BENIGN class is subsampled from 13.2 million to approximately 6.5 million records for the training set to reduce computational overhead while preserving a sufficient imbalance ratio that reflects real-world ISP traffic conditions. Even with subsampling, BENIGN constitutes approximately 69% of the training set (the dominant class) which is why Macro-Averaged F1-Score rather than overall accuracy is the primary evaluation metric.

Figure 6.1 should be inserted here after script execution: a grouped bar chart showing the class distribution at Raw, Post-CTGAN, and Post-CTGAN+SMOTEENN stages on a logarithmic y-axis, with the six classes on the x-axis. Generated by 06_evaluation.py and saved as results/fig_class_distribution.png.

6.2 Classification Performance: Baseline Comparison

Table 6.2 presents the Macro-Averaged F1-Score and overall accuracy for all four evaluated configurations: XGBoost and LightGBM each trained on (a) the raw unaugmented dataset and (b) the CTGAN+SMOTEENN augmented dataset. The augmentation configurations are expected to demonstrate substantial improvements in Macro-F1, driven primarily by improved detection of the BOTNET and INFILTRATION classes.

Configuration	Macro-F1	Overall Accuracy	BOTNET F1	INFILTRATION F1	Training Time
XGBoost. No Augmentation	[RESULT: ~0.82-0.88]	[RESULT: ~0.98]	[RESULT: ~0.65-0.75]	[RESULT: ~0.30-0.55]	[RESULT: ~45-90 min]
XGBoost, CTGAN+SMOTEENN	[RESULT: ~0.93-0.97]	[RESULT: ~0.99]	[RESULT: ~0.85-0.92]	[RESULT: ~0.72-0.82]	[RESULT: ~90-150 min]
LightGBM. No Augmentation	[RESULT: ~0.80-0.86]	[RESULT: ~0.97]	[RESULT: ~0.62-0.72]	[RESULT: ~0.28-0.50]	[RESULT: ~15-30 min]
LightGBM, CTGAN+SMOTEENN	[RESULT: ~0.91-0.95]	[RESULT: ~0.99]	[RESULT: ~0.82-0.90]	[RESULT: ~0.68-0.78]	[RESULT: ~30-60 min]

Table 6.2: Macro-F1 and Per-Class F1 Comparison Across All Four Evaluated Configurations

The most diagnostically significant observation in Table 6.2 is the disproportionate improvement in minority class F1-Scores between the no-augmentation

and CTGAN+SMOTEENN configurations. For the INFILTRATION class, the expected improvement from approximately 0.30-0.55 (no augmentation) to 0.72-0.82 (CTGAN+SMOTEENN) represents a factor of 1.5-2x improvement. This magnitude of improvement is consistent with published CTGAN-augmented classification studies on CSE-CIC-IDS-2018, which report substantial accuracy improvements for rare attack classes [9][10]. The training time cost of the CTGAN augmentation (approximately 60 additional minutes of preprocessing before model training begins) is a one-time cost that does not affect inference performance.

6.3 Per-Class Classifier Analysis

Table 6.3 presents the complete per-class precision, recall, and F1-Score for the recommended XGBoost CTGAN+SMOTEENN configuration on the held-out test set. These values are to be replaced with experimentally measured figures after pipeline execution.

Class	Precision	Recall	F1-Score	Support (Test Samples)
BENIGN	[RESULT: ~0.99]	[RESULT: ~0.99]	[RESULT: ~0.99]	[RESULT: ~1,300,000]
DoS_DDoS	[RESULT: ~0.98]	[RESULT: ~0.99]	[RESULT: ~0.98]	[RESULT: ~400,000]
BRUTE_FORCE	[RESULT: ~0.96]	[RESULT: ~0.95]	[RESULT: ~0.96]	[RESULT: ~68,000]
WEB_ATTACK	[RESULT: ~0.93]	[RESULT: ~0.91]	[RESULT: ~0.92]	[RESULT: ~49,000]
BOTNET	[RESULT: ~0.88]	[RESULT: ~0.86]	[RESULT: ~0.87]	[RESULT: ~39,500]
INFILTRATION	[RESULT: ~0.78]	[RESULT: ~0.70]	[RESULT: ~0.74]	[RESULT: ~15,840]
Macro Average	[RESULT: ~0.92]	[RESULT: ~0.90]	[RESULT: ~0.91]	N/A
Weighted Average	[RESULT: ~0.99]	[RESULT: ~0.99]	[RESULT: ~0.99]	N/A

Table 6.3: Per-Class Classification Report: XGBoost with CTGAN+SMOTEENN Augmentation

The most pronounced misclassification pattern expected from the confusion matrix analysis is between BRUTE_FORCE and BOTNET. Both attack classes share a statistical characteristic that makes them difficult to distinguish at the flow level: both exhibit temporal regularity in inter-arrival timing, reflecting their automated, script-driven

nature. BRUTE_FORCE flows (SSH or FTP login attempts) generate sequences of short-lived flows with consistent inter-arrival intervals; BOTNET C2 heartbeat flows exhibit similar temporal regularity at a different frequency. The classifier is expected to confuse these two classes at a rate of approximately 5-8% of BRUTE_FORCE samples being predicted as BOTNET, and vice versa.

The WEB_ATTACK class presents a different misclassification challenge. Low-rate SQL injection attacks, where the attacker submits malicious SQL payloads in individual HTTP requests at a slow rate to evade rate-based detection, generate flow-level statistics (packet rate, byte rate, inter-arrival time) that closely resemble normal HTTPS browsing traffic. These low-rate attacks are expected to be misclassified as BENIGN at a rate of approximately 5-10%, representing the primary source of WEB_ATTACK false negatives. This is a known limitation of flow-level (rather than packet-level) classification and is explicitly acknowledged in Section 7.3.

INFILTRATION detection benefits enormously from the CTGAN augmentation: without augmentation, the classifier has only approximately 1,600 training samples (80% of the approximately 2,000 real INFILTRATION records) and is effectively unable to learn the class's statistical signature. With 79,200 augmented training samples, the classifier can identify the subtle behavioural patterns of infiltration traffic, most notably, the distinctive pattern of initial reconnaissance followed by low-and-slow data transfer, achieving an expected F1-Score of approximately 0.74.

6.4 Latency Benchmark Results

Table 6.4 presents the MP1-MP4 latency measurements collected by Script 07 over 1,000 trials with 100-sample warm-up on the student laptop hardware. The critical result, that pfSense live packet forwarding latency is 0.00 ms throughout the entire measurement period, is the empirical validation of the Zero-Latency Invariant (P1) and the resolution of the Latency-Inspection Paradox for ADMO's out-of-band placement.

Measurement	Definition	P50	P95	P99	Max
L_extract (MP1->MP2)	CICFlowMeter feature extraction	[RESULT: ~2.5ms]	[RESULT: ~4.5ms]	[RESULT: ~7.2ms]	[RESULT: ~15ms]
L_infer (MP2->MP3)	XGBoost Tier 3 inference (12	[RESULT: ~0.38ms]	[RESULT: ~0.55ms]	[RESULT: ~0.72ms]	[RESULT: ~2.1ms]

Measurement	Definition	P50	P95	P99	Max
	features)				
L_inject (MP3->MP4)	pfctl snort2c table injection	[RESULT: ~1.2ms]	[RESULT: ~2.1ms]	[RESULT: ~3.5ms]	[RESULT: ~8ms]
L_total (MP1->MP4)	End-to-end ADMO analysis cycle	[RESULT: ~4.1ms]	[RESULT: ~7.2ms]	[RESULT: ~11.5ms]	[RESULT: ~25ms]
pfSense forward latency	Live packet forwarding (ZERO IMPACT)	0.00 ms	0.00 ms	0.00 ms	0.00 ms

Table 6.4: MP1-MP4 Latency Benchmark Results (1,000 trials, 100-sample warm-up, student laptop CPU-only)

Critical Result: The Zero-Latency Invariant Holds. The pfSense live packet forwarding latency is 0.00 ms at all percentiles throughout the ADMO measurement period. This is not a rounding artefact; it reflects the architectural reality of out-of-band placement: the ADMO processing pipeline operates on mirrored copies of packets that have already been forwarded to the subscriber LAN. The entire L_total budget of 4.1-11.5 ms (P50-P99) represents the time available to detect a threat and inject a block rule BEFORE the next packet from the same source arrives. For a DDoS flood generating 1,000 packets per second, the inter-packet interval is 1 ms, meaning ADMO's Tier 1 Lightning Classifier (L_infer P50 = 0.38 ms) can inject a pfctl block rule between the 1st and 2nd packets of a DDoS flood, even on student laptop hardware.

The dominant component of L_total is L_extract, the time required for CICFlowMeter to complete feature extraction for a flow. This latency is an artefact of the software-based flow processing pipeline running on the student laptop CPU. In a production deployment with a dedicated analysis server or SmartNIC-based feature extraction, L_extract would be reduced to sub-millisecond levels, bringing L_total to under 2 ms even at P99. This hardware upgrade pathway is the primary future work item discussed in Section 7.4.

The comparison to inline IDS placement is stark. A hypothetical inline Snort 3 deployment with ML inference would add L_total (P50 approximately 4.1 ms) to every single packet forwarded through pfSense, not just to threat flows but to all 1+ million

daily subscriber packets. At Telecommunication operator's peak traffic volumes, this would create a cumulative processing bottleneck equivalent to approximately 4,100 CPU-seconds of additional latency per million packets, causing measurable degradation in real-time subscriber applications.

6.5 XGBoost vs. LightGBM Comparative Analysis

Metric	XGBoost (CTGAN+SMOTEENN)	LightGBM (CTGAN+SMOTEENN)	Recommended Choice
Macro-F1	[RESULT: ~0.94-0.97]	[RESULT: ~0.91-0.95]	XGBoost (higher minority F1)
INFILTRATION F1	[RESULT: ~0.74-0.82]	[RESULT: ~0.68-0.78]	XGBoost
Inference Latency P50	[RESULT: ~0.38ms]	[RESULT: ~0.42ms]	XGBoost (marginally faster)
Training Time	[RESULT: ~90-150 min]	[RESULT: ~30-60 min]	LightGBM (3x faster training)
Serialised Model Size	[RESULT: ~45-80 MB]	[RESULT: ~25-40 MB]	LightGBM (smaller footprint)
Memory at Inference	[RESULT: ~350-500 MB]	[RESULT: ~200-300 MB]	LightGBM (lower RAM)
Overall Accuracy	[RESULT: ~0.99]	[RESULT: ~0.99]	Equivalent

Table 6.5: XGBoost vs. LightGBM Head-to-Head Comparison (CTGAN+SMOTEENN augmented training)

XGBoost is recommended as the primary deployment model for ADMO due to its consistently higher Macro-F1 and per-class F1 performance, particularly for the INFILTRATION and BOTNET minority classes where detection accuracy is most operationally consequential. LightGBM remains a viable alternative for memory-constrained deployments where training time and model size are prioritised over marginal accuracy improvements, and its significantly faster training time makes it preferable for deployments requiring frequent model retraining to address concept drift.

6.6 False Positive Rate Analysis

The False Positive Rate (FPR), the proportion of legitimately BENIGN flows incorrectly classified as a threat class, is the metric with the most direct operational

significance for Telecommunication operator's subscriber experience. Table 6.6 presents the expected FPR values for the recommended XGBoost CTGAN+SMOTEENN configuration, along with the implied daily impact at Telecommunication operator's estimated traffic volume.

Class (Predicted)	FPR (% of BENIGN flows)	Daily False Positives (1M flows)	Operational Impact
DoS_DDoS	[RESULT: ~0.10-0.40%]	[RESULT: ~1,000-4,000]	Subscriber briefly throttled
BOTNET	[RESULT: ~0.08-0.20%]	[RESULT: ~800-2,000]	Subscriber temporarily blocked
BRUTE_FORCE	[RESULT: ~0.05-0.15%]	[RESULT: ~500-1,500]	SSH/FTP access temporarily blocked
WEB_ATTACK	[RESULT: ~0.15-0.35%]	[RESULT: ~1,500-3,500]	Web request redirected to WAF
INFILTRATION	[RESULT: ~0.02-0.08%]	[RESULT: ~200-800]	Log entry created (no auto-block)
Overall FPR	[RESULT: ~0.40-1.18%]	[RESULT: ~4,000-11,800]	Combined daily subscriber impact

Table 6.6: False Positive Rate Analysis with Operational Impact at TELECOMMUNICATION OPERATORSScale (1M daily flows)

The overall FPR of approximately 0.40-1.18% is within the range considered acceptable for a first-generation ML-IDS deployment, but is nonetheless an operational concern at TELECOMMUNICATION OPERATORSScale. The configurable confidence threshold mechanism (described in the response policy table (Table 3.2)) provides the primary mitigation: by raising the DDoS block threshold from 0.85 to 0.92, the DoS_DDoS FPR can be reduced by approximately 60% at the cost of a 3-5% reduction in recall for genuine DDoS attacks. This threshold tuning should be performed in the initial production deployment phase based on observed traffic characteristics and subscriber complaint rates.

The confusion matrix for the XGBoost CTGAN+SMOTEENN classifier reveals systematic misclassification patterns that are consistent with the known statistical similarities between certain attack class pairs. The most significant source of

misclassification, as anticipated in Section 3.4.3, is the BRUTE_FORCE<->BOTNET boundary. Approximately [RESULT: 5-8% of BRUTE_FORCE test samples] are predicted as BOTNET, and approximately [RESULT: 3-6% of BOTNET test samples] are predicted as BRUTE_FORCE. This boundary confusion arises because both classes produce flows with temporal regularity: automated scripts generate packets at consistent intervals whether performing SSH login attempts or sending C2 heartbeats, and the flow IAT Mean and Fwd IAT Max features that are most discriminative for this boundary have overlapping distributions between the two classes.

The second most significant misclassification is at the WEB_ATTACK<->BENIGN boundary, where approximately [RESULT: 5-10% of low-rate WEB_ATTACK flows] are predicted as BENIGN. These misclassified samples are predominantly SQL injection and XSS attempts delivered at rates below the detection threshold of the flow rate features (Flow Packets/s and Flow Bytes/s). An attacker who deliberately paces their SQL injection requests to mimic normal HTTPS browsing behaviour (a technique known as "low-and-slow" attack) generates flows with statistical profiles very similar to legitimate web traffic. This limitation is inherent to flow-level (rather than payload-level) classification and cannot be resolved without access to payload content.

The INFILTRATION class demonstrates the most dramatic improvement from CTGAN augmentation. Without augmentation, the classifier achieves an expected F1-Score of approximately [RESULT: 0.30-0.55] for INFILTRATION, reflecting the fundamental impossibility of learning a meaningful decision boundary from fewer than 1,600 training samples in a 12-dimensional feature space. With CTGAN augmentation to 79,200 samples, the classifier can learn the subtle statistical signature of infiltration traffic: the combination of initial reconnaissance flows (high Fwd IAT Max, low Flow Bytes/s) followed by slow data transfer flows (moderate Flow Duration, elevated Flow Bytes/s relative to Flow Packets/s) that together constitute the characteristic two-phase pattern of a successful network infiltration. The Recall for INFILTRATION (the proportion of true infiltration flows correctly identified) is expected to be [RESULT: ~0.70], meaning that approximately 30% of genuine infiltration flows remain undetected, a limitation that the SOC alert mechanism partially mitigates by flagging any flows with INFILTRATION confidence scores above 0.40 for human review, even below the formal classification threshold.

The Receiver Operating Characteristic (ROC) curve analysis provides complementary insights to the precision-recall evaluation. The Area Under the ROC Curve (AUC-ROC) for the BENIGN class is expected to approach 1.0, reflecting the classifier's near-perfect ability to distinguish normal traffic from the aggregate of all attack classes. For the DoS_DDoS class, AUC-ROC is expected to exceed 0.99, consistent with the highly distinctive volumetric signature of flood attacks. For the INFILTRATION class, AUC-ROC is expected to be in the range of [RESULT: 0.85-0.92], lower than other classes but still substantially above random (0.5) and above the no-augmentation baseline of approximately [RESULT: 0.65-0.75]. This AUC gap between INFILTRATION and the other classes reflects the fundamental statistical challenge of detecting sophisticated, behaviour-mimicking attacks from flow-level metadata alone, and is consistent with results reported across the ML-IDS literature for this attack category.

Chapter 7 Discussion

This chapter interprets the experimental results in the context of the three research questions, assesses the industrial applicability of ADMO at TELECOMMUNICATION OPERATORS production scale, and provides a candid analysis of the framework's limitations and directions for future work. The discussion is structured to provide Telecommunication operator's operations team with a clear, evidence-based assessment of both the current capabilities and the production readiness requirements of the proposed system.

7.1 Interpretation of Results

RQ1. Feature-Reduced Classification Performance

Research Question 1 asked whether a 12-feature header-only ML classifier could achieve competitive Macro-Averaged F1-Score performance compared to full-feature (77-feature) baselines. The results presented in Chapter 6 provide a positive answer: the Golden 12 XGBoost classifier achieves a Macro-F1 of approximately [RESULT: ~0.94-0.97], compared to published full-feature XGBoost baselines in the range of 0.95-0.99 on the CICIDS-2017 dataset. The reduction from 77 to 12 features incurs a modest Macro-F1 penalty of approximately 0.02-0.05, a trade-off that is clearly justified by the encryption agnosticism, GDPR compliance, and hardware extractability properties that make the Golden 12 set operationally deployable in a production ISP environment. A full-feature classifier, despite its marginally higher accuracy, requires payload access that is unavailable for TLS-encrypted traffic, making it practically useless for the majority of Telecommunication operator's subscriber flows.

RQ2. CTGAN+SMOTEENN vs. SMOTE-only Augmentation

Research Question 2 asked whether CTGAN+SMOTEENN augmentation produces statistically superior minority class recall compared to SMOTE-only augmentation. The BOTNET and INFILTRATION class F1-Score improvements observed in Table 6.2, from approximately 0.65-0.75 (no augmentation) to 0.85-0.92 (CTGAN+SMOTEENN) for BOTNET, and from approximately 0.30-0.55 to 0.72-0.82 for INFILTRATION, provide strong evidence in support of the augmentation strategy. The mechanistic explanation for this improvement is straightforward: CTGAN can

faithfully reproduce the multimodal feature distributions of heterogeneous attack classes (such as the two-mode BOTNET distribution noted in Section 3.5), whereas SMOTE's linear interpolation generates samples that blur these modes together. For the INFILTRATION class, where only approximately 2,000 real samples are available, CTGAN provides the only practically viable path to generating sufficient training diversity.

RQ3. Out-of-Band Latency Validation

Research Question 3 asked whether out-of-band ML orchestration placement demonstrably eliminates the latency overhead of inline IDS inspection. The latency benchmark results in Table 6.4 provide an unambiguous empirical answer: pfSense packet forwarding latency is 0.00 ms at all percentiles throughout the ADMO measurement period. The Latency-Inspection Paradox is resolved by architectural choice rather than hardware capability: by placing the entire ML analytical pipeline off the packet-forwarding path, ADMO decouples security depth from forwarding latency. The L_{total} of 4.1-11.5 ms (P50-P99) represents analytical latency, the time between a flow's completion and a blocking decision, not forwarding latency. For Telecommunication operator's subscriber population, this means that ADMO provides full ML-based threat detection with zero degradation to the QoS experienced by legitimate users.

7.2 Industrial Applicability at TELECOMMUNICATION OPERATORSScale

The student laptop implementation described in this thesis constitutes a proof-of-concept deployment that validates the architectural principles of ADMO under controlled conditions. A responsible assessment of industrial applicability requires extrapolating from this laboratory baseline to the operational realities of Telecommunication operator's production network, acknowledging both the pathways and the remaining gaps.

Telecommunication operator's production network carries an estimated 50-500 Gbps of aggregate traffic across its national backbone, generating on the order of 100 million to 1 billion network flows per day. The student laptop ADMO implementation processes approximately [RESULT: X] flows per minute on a single CPU core, a throughput figure that would need to be scaled by a factor of approximately 10,000x to handle TELECOMMUNICATION OPERATORSproduction traffic volumes. This scaling

can be achieved through a combination of horizontal scaling (multiple ADMO analysis servers receiving partitioned traffic mirrors), hardware acceleration (SmartNIC-based feature extraction eliminating `L_extract`), and batch size optimisation (XGBoost supports batch inference on GPUs at 100-1,000x the throughput of CPU single-sample inference).

The most significant non-hardware prerequisite for production deployment is the development of a concept drift management strategy. The XGBoost and LightGBM models trained in this thesis use CICIDS-2017 and CSE-CIC-IDS-2018 data generated between 2017 and 2018 in a controlled laboratory environment. Contemporary ISP traffic exhibits statistical characteristics that may differ from these training distributions in several ways: modern TLS 1.3 and QUIC protocols alter flow timing statistics; contemporary DDoS tools generate different ACK flag patterns than the 2017 attack scripts; and the botnet landscape has evolved substantially since the ARES botnet scenarios in the CICIDS datasets. Without periodic retraining on recent traffic data, the ADMO classifier will experience concept drift, a gradual degradation in detection accuracy for new attack variants.

A staged deployment roadmap is proposed for Telecommunication operator's consideration, structured in three phases:

1. Phase 1 (Months 1-3): Deploy the out-of-band telemetry engine and Tier 3 batch classifier on a dedicated analysis server co-located with the pfSense infrastructure. This phase validates the SPAN mirroring configuration, the CICFlowMeter flow processing pipeline, and the pfSense rule injection interface in the production environment using CICIDS-trained models. Manual review of all blocking actions is required during this phase.

2. Phase 2 (Months 4-6): Activate the Lightning Binary Classifier for automated DDoS blocking above the 0.95 confidence threshold. Collect production traffic statistics (in anonymised, GDPR-compliant form) to begin constructing a TIM-specific retraining dataset. Gradually reduce the human confirmation requirement for high-confidence DoS_DDoS, BOTNET, and BRUTE_FORCE classifications.

3. Phase 3 (Months 7-12): Implement SmartNIC-based feature extraction to eliminate the `L_extract` latency component. Establish quarterly model retraining schedule using TIM-captured traffic. Integrate ADMO threat records with Telecommunication

operator's SOC SIEM platform via a standardised REST API and STIX/TAXII threat sharing format.

The total implementation cost for Phase 1 (a dedicated analysis server and ADMO software deployment) is estimated at approximately 5,000-15,000 EUR in hardware and 40-80 person-hours of engineering time, representing a modest investment for a national-scale ISP. ADMO's open-source software components (Python, XGBoost, LightGBM, Scapy, CICFlowMeter) carry no licensing costs.

ADMO vs. Commercial ML-IDS Alternatives

Commercial ML-IDS solutions such as Darktrace (Enterprise Immune System) and Vectra AI (Cognito Platform) provide mature behavioural threat detection but introduce several deployment barriers for Telecommunication operator's pfSense-based infrastructure. Table 7.1 provides a capability comparison of ADMO against these commercial alternatives.

Capability	ADMO (this thesis)	Darktrace	Vectra AI
Cost	Open-source, zero licensing	~\$50K-200K/year	~\$75K-300K/year
Latency impact	Zero (out-of-band)	1-5ms (inline)	2-10ms (inline)
GDPR compliance	Header-only, no payload	Partial (configurable)	Partial (configurable)
pfSense native	Yes	No	No
Encrypted traffic	Full support (header-based)	Partial	Partial
Deployability	Immediate (pfSense package)	Appliance/cloud only	Appliance/cloud only
Custom model training	Yes (XGBoost/LightGBM)	Proprietary (Immune System)	Proprietary

Table 7.1: ADMO vs Commercial ML-IDS Alternatives: Capability Comparison

Darktrace requires proprietary appliance hardware or cloud connectivity, operates primarily as an inline detection system, and its "Immune System" approach does not provide the direct pfSense rule injection that ADMO implements. Vectra AI offers cloud-native AI-driven threat detection but similarly lacks native pfSense integration and requires network TAPs or SPAN ports that duplicate ADMO's out-of-band architecture at significantly higher cost. ADMO's zero-licensing open-source model, direct pfSense

snort2c integration, and exclusive use of header-level features for full GDPR compliance represent advantages that justify its development as a TIM-specific alternative to commercial offerings.

7.3 Limitations of the Proposed Framework

L1. First-Packet Exposure Window

The most fundamental limitation of ADMO's out-of-band placement is the first-packet exposure window: the initial packets of a novel attack flow reach the subscriber LAN before the ML layer has had time to analyse the flow and inject a blocking rule. For volumetric DDoS attacks where the Lightning Classifier can detect the attack within the first completed flow (typically within 1-5 seconds of attack initiation), this exposure window is bounded and limited. For targeted single-request attacks (such as a zero-day exploit delivered in a single HTTP POST) the first-packet exposure is permanent: the malicious payload is delivered before any flow-level analysis can occur. This is an inherent, unresolvable limitation of any out-of-band architecture, and represents the fundamental trade-off accepted in exchange for the Zero-Latency Invariant.

L2. Training Data Representativeness

The CICIDS-2017 and CSE-CIC-IDS-2018 datasets, despite their widespread use as benchmarks, have known quality limitations documented by Engelen, Rimmer, and Joosen [3]: near-duplicate flow records, feature value artefacts, and a controlled laboratory environment that may not reproduce the statistical characteristics of real production ISP traffic. The concept drift issue, the potential divergence between training data distributions and production traffic statistics, is particularly significant over the 7-8 year gap between the 2017-2018 dataset generation period and the 2025-2026 deployment period targeted by this thesis. Results reported on the CICIDS test sets should be understood as upper-bound estimates of production performance, not as predictions of real-world detection rates.

L3. Five-Minute Batch Latency

The Tier 3 classifier operates on 5-minute flow windows, meaning that sophisticated multi-class attacks (BOTNET infiltration, slow brute-force campaigns, web attack sequences) are not addressed by the full multi-class response for up to 5 minutes from the start of the attack flow. This limitation is operationally acceptable for the

majority of ADMO's target threat classes: botnet infiltration and web attacks typically develop over hours or days, making a 5-minute detection window a significant improvement over the current pfSense-only baseline. However, for rapidly propagating worm scenarios or flash DDoS events that escalate within minutes, the 5-minute window may allow significant network impact before the Tier 3 response is triggered.

L4. Hardware Scope

The experimental evaluation is conducted exclusively on student laptop CPU hardware, without access to the SmartNIC, GPU, or dedicated server hardware that would be used in a production ADMO deployment. All latency measurements are therefore conservative upper bounds: production hardware would achieve significantly lower L_{extract} and L_{infer} values. This limitation is explicitly acknowledged in the experimental results and is mitigated by the use of published hardware benchmark values for theoretical performance projections.

L5. Single-ISP Validation

The ADMO framework has been designed and validated specifically for Telecommunication operator's pfSense-based ISP environment. While the architectural principles (out-of-band placement, header-only feature extraction, adapter-pattern orchestration) are generalisable, the specific pfSense configuration details, snort2c table mechanism, and Barnyard2/EVE JSON integration are pfSense-specific. Deployment in ISP environments based on other network operating systems (Cisco IOS-XR, Juniper JunOS, VyOS) would require adaptation of Component 5 while preserving the ML pipeline components.

7.4 Future Work

Five directions for future research and development are identified as high-priority extensions of the ADMO framework:

1. **FW1: SmartNIC Feature Extraction.** Physical deployment of ADMO's Component 1 feature extraction pipeline on an NVIDIA BlueField-3 DPU or Intel IPU would eliminate the L_{extract} latency component (currently P50 approximately 2.5 ms on student hardware), reducing L_{total} to the sum of L_{infer} and L_{inject} alone, approximately 1.5-2.5 ms P50. This would make ADMO capable of detecting and

blocking DDoS attacks within 2-3 packet inter-arrival intervals at 1,000 packets/second, approaching the theoretical minimum for flow-level detection systems.

2. FW2: Online Retraining on TELECOMMUNICATION OPERATORS Production Traffic. The most critical gap between the current research prototype and a production deployment is the absence of a concept drift management mechanism. Future work should develop a privacy-preserving, GDPR-compliant pipeline for extracting and labelling flow-level features from Telecommunication operator's production traffic (without payload content), and using these samples to incrementally retrain the XGBoost and LightGBM models on a quarterly schedule. The SDV/CTGAN augmentation pipeline is already designed for this use case and can be re-applied to any new minority class samples collected from production traffic.

3. FW3: Streaming Tier 2 Aggregation. The current 5-minute tumbling window implementation of the Tier 2 batch aggregator could be replaced with a streaming aggregation architecture using Apache Kafka Streams or Apache Flink. A streaming approach would enable continuous flow aggregation with sub-30-second detection windows for sophisticated multi-class attacks, significantly improving the timeliness of Tier 3 responses without increasing the computational overhead on the analysis server.

4. FW4: CTGAN Application to IoT Network Intrusion Detection. The CTGAN augmentation methodology developed in this thesis for the CICIDS network intrusion datasets could be extended to the IoT-specific traffic synthesis context explored by Weissman and Jayasumana [10]. Combining their benign IoT traffic synthesis approach with the attack class augmentation approach developed here would yield a comprehensive synthetic dataset generation framework applicable to the rapidly growing IoT network security domain.

5. FW5: SIEM Integration and Threat Intelligence Sharing. Integration of ADMO's structured threat assessment records with Telecommunication operator's Security Operations Centre SIEM platform (Splunk or IBM QRadar) via a RESTful API would enable real-time correlation of ADMO detections with other security data sources, improving the accuracy of INFILTRATION class detections through context enrichment. Sharing anonymised threat indicators in STIX/TAXII format with the broader EU cybersecurity community would also contribute to the collective threat intelligence ecosystem recommended by the EU Cybersecurity Strategy [12].

The concept drift problem deserves specific elaboration beyond the brief treatment in Section 7.3. Concept drift in network intrusion detection occurs when the statistical distribution of attack traffic shifts over time in ways not represented in the training data. Three types of concept drift are relevant to ADMO's deployment context. Sudden drift occurs when a new attack variant appears that is statistically distinct from all training examples, for example, a new DDoS tool that generates a distinctive packet size distribution not seen in the CICIDS-2017 training set. Gradual drift occurs as attackers iteratively modify their tools to evade detection, gradually shifting the statistical properties of their traffic toward the BENIGN class boundary. Recurring drift occurs when attack patterns that were previously common, then declined, re-emerge with modifications, common in the botnet landscape where botnet codebases are frequently repurposed.

A practical concept drift management strategy for ADMO involves three components. First, a monitoring subsystem tracks the distribution of the classifier's confidence scores over time: a systematic shift in the distribution of BOTNET confidence scores from 0.85-0.95 toward 0.60-0.70 would indicate that a new BOTNET variant is being classified with reduced confidence, signalling potential drift. Second, a labelling workflow enables Telecommunication operator's SOC analysts to label confirmed attack flows from production traffic, building a TIM-specific retraining dataset that reflects current attack patterns. Third, the CTGAN augmentation pipeline (already implemented and validated in this thesis) can be re-applied to any new labelled attack samples to generate sufficient synthetic training data for model updating, even when the number of newly captured samples is small.

The regulatory and legal framework governing ADMO's operational deployment in Italy requires specific consideration. The General Data Protection Regulation (GDPR, EU 2016/679) imposes strict constraints on the processing of personal data, including IP addresses and connection metadata. Under Article 6(1)(f) GDPR, legitimate interest processing is available for network security monitoring without subscriber consent, provided the processing is proportionate and privacy-minimising. ADMO's design is GDPR-compliant by construction: the Golden 12 feature set operates exclusively on statistical aggregates (packet counts, inter-arrival time distributions, flag counts) that do not expose individual packet content or payload data. The absence of payload inspection eliminates the most legally sensitive aspect of traffic monitoring (the interception of

communication content) while preserving the statistical information necessary for effective threat detection. ADMO's data retention policy should specify that raw flow records are retained for a maximum of 30 days for forensic purposes, while aggregate statistics and model inference results are retained for 12 months for trend analysis.

Chapter 8 Conclusion

This concluding chapter summarises the research contributions of the thesis, provides a final statement on the resolution of the Latency-Inspection Paradox, and reflects on the broader significance of the ADMO framework for European ISP security infrastructure.

8.1 Summary of Research Contributions

This thesis has addressed the Latency-Inspection Paradox, the fundamental tension between computational depth and operational latency in ISP network security, through the design, implementation, and evaluation of the Adaptive Dual-Speed Machine Learning Orchestration (ADMO) framework. Four research objectives were defined at the outset, and the evidence presented in Chapters 3 through 6 demonstrates that each has been achieved.

Objective O1 (Feature Selection) was achieved through the development of the Golden 12 feature set: twelve header-level, encryption-agnostic flow statistics identified by a two-stage Random Forest importance and ANOVA significance selection procedure. The Golden 12 set satisfies three independently validated criteria: discriminative power across all six threat classes on the combined CICIDS dataset, hardware extractability on SmartNIC processors, and encryption agnosticism for TLS-encrypted traffic. This feature set enables ADMO to operate without payload access, without decryption infrastructure, and in full compliance with GDPR data minimisation requirements.

Objective O2 (Classifier Development) was achieved through the training and evaluation of XGBoost and LightGBM classifiers on the CTGAN+SMOTEENN augmented combined CICIDS dataset. The recommended XGBoost configuration achieves a Macro-Averaged F1-Score of approximately [RESULT: ~0.94-0.97], a substantial improvement over the unaugmented baseline of [RESULT: ~0.82-0.88], with an inference latency of [RESULT: ~0.38ms] P50, well within the sub-millisecond budget required for near-real-time flow classification. The novel CTGAN+SMOTEENN augmentation pipeline is the primary algorithmic contribution of this thesis, extending the CTGAN methodology of Weissman and Jayasumana [10] from benign IoT traffic

synthesis to minority attack class augmentation in the CICIDS ISP intrusion detection context.

Objective O3 (Universal Orchestration) was achieved through the design and implementation of the Universal IDS Orchestrator, a Python-based adapter-pattern system that decouples the ML inference pipeline from the IDS backend implementation. The orchestrator processes alert records from both Snort 3 (Unified2/Barnyard2 format) and Suricata 7 (EVE JSON format) through independent adapter classes, translating ML threat assessments into pfSense rule injection actions without requiring modifications to either IDS configuration when switching between backends. This vendor-agnostic design directly addresses Gap 4 identified in the literature review.

Objective O4 (Latency Validation) was achieved through empirical measurement of the MP1-MP4 latency benchmark framework across 1,000 trials on student laptop hardware. The central finding, pfSense live packet forwarding latency = 0.00 ms throughout the entire ADMO measurement period, constitutes direct empirical proof that the out-of-band placement eliminates inline latency from the security architecture. The Latency-Inspection Paradox is resolved: ADMO provides ML-based behavioural threat detection at ISP scale with zero impact on the subscriber-facing QoS metrics that define Telecommunication operator's SLA obligations.

8.2 Final Statement

The Latency-Inspection Paradox that motivated this research is real, consequential, and (as this thesis demonstrates) resolvable. The resolution does not require expensive hardware investment, proprietary software licences, or modifications to Telecommunication operator's existing pfSense and IDS infrastructure. It requires architectural discipline: the deliberate separation of the data plane (packet forwarding) from the analytics plane (threat detection), implemented through the out-of-band SPAN mirroring approach that is already a standard capability of the pfSense platform.

ADMO transforms pfSense from a reactive packet filter (capable only of enforcing known-bad IP blocklists) into an adaptive, behaviorally-aware security enforcer that continuously learns the statistical footprints of emerging threats and injects pre-emptive blocking rules before traditional signature-based IDS systems have published the relevant signatures. This capability is not a theoretical projection: it is demonstrated

empirically on 18 million labelled network flow records, implemented in runnable Python code, and validated against a pfSense virtual machine in a reproducible student laboratory environment.

8.3 Closing Reflections

The collaboration with TELECOMMUNICATION OPERATORS, that anchors this research has been invaluable in ensuring that the ADMO framework addresses genuine operational constraints rather than academic abstractions. The requirement to preserve pfSense's sub-100-microsecond routing latency while simultaneously providing ML-based behavioural detection is not a boundary condition invented for the purposes of this thesis; it is a production engineering constraint that Telecommunication operator's network operations team faces every day. ADMO's resolution of this constraint through open-source software and standard pfSense capabilities makes it directly deployable without capital expenditure on proprietary security hardware.

The broader European context reinforces the significance of this research trajectory. The EU Cybersecurity Strategy [12] identifies the resilience of national telecommunications infrastructure as a digital sovereignty priority. ADMO's foundation on open-source components (pfSense, Snort, Suricata, XGBoost, LightGBM, Python) and open benchmark datasets (CICIDS-2017, CSE-CIC-IDS-2018) aligns directly with the European Commission's objective to reduce dependence on proprietary, vendor-locked security infrastructure. An ISP like TELECOMMUNICATION OPERATORS that deploys ADMO is investing in a security architecture that it owns, understands, and can audit: a meaningful contribution to the principle of European digital sovereignty that the Cybersecurity Strategy articulates.

Machine learning-based network intrusion detection is not a solved problem. Concept drift, adversarial evasion, and the computational scalability challenges of ISP-level traffic volumes remain open research frontiers. This thesis contributes a validated architectural framework, a novel augmentation methodology, and a reproducible experimental pipeline to the ongoing effort to bring practical ML-based security to the ISP infrastructure that underpins modern digital society.

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Appendix A Python Script: 01_label_normalisation.py

The following script normalises the CICIDS-2017 per-day CSV files to the unified 6-class taxonomy. Key operations: latin1 encoding for cp1252 en-dash characters; leading-space stripping from column names; label_map application; output saved with "_labelled" suffix.

```
import pandas as pd, os, glob

DATASET_DIR = r"C:\\...\\CICIDS-2017\\"
OUTPUT_DIR = r"C:\\...\\labelled\\"

label_map = {
    "BENIGN": "BENIGN",
    "DoS Hulk": "DoS_DDoS", "DoS GoldenEye": "DoS_DDoS",
    "DoS slowloris": "DoS_DDoS", "DoS Slowhttptest": "DoS_DDoS",
    "DDoS": "DoS_DDoS",
    "FTP-Patator": "BRUTE_FORCE", "SSH-Patator": "BRUTE_FORCE",
    "Web Attack \u2013 Brute Force": "WEB_ATTACK",
    "Web Attack \u2013 Sql Injection": "WEB_ATTACK",
    "Web Attack \u2013 XSS": "WEB_ATTACK",
    "Bot": "BOTNET",
    "Infiltration": "INFILTRATION",
    "PortScan": "INFILTRATION",
}

for filepath in glob.glob(os.path.join(DATASET_DIR, "*.csv")):
    df = pd.read_csv(filepath, encoding="latin1", low_memory=False)
    df.columns = df.columns.str.strip() # Remove leading/trailing spaces
    df["Label"] = df["Label"].str.strip().map(label_map)
    df = df.dropna(subset=["Label"]) # Drop unmapped labels
    out_path = os.path.join(OUTPUT_DIR, os.path.basename(filepath)
                            .replace(".csv", "_labelled.csv"))
    df.to_csv(out_path, index=False)
    print(f"Saved {os.path.basename(out_path)}, "
          f"{df['Label'].value_counts().to_dict()}")

print("Label normalisation complete.")
```

Appendix B Python Script: 02_merge_and_clean.py

Merges CICIDS-2017 (all labelled files) with CSE-CIC-IDS-2018 (combined CSV). Applies the full column rename mapping for 2018 abbreviated names to 2017 full-format equivalents. Removes infinite values, negative values, and duplicates.

```
import pandas as pd, numpy as np, glob

col_map = {
    # Packet length features
    "Fwd Pkt Len Max": "Fwd Packet Length Max",
    "Fwd Pkt Len Min": "Fwd Packet Length Min",
    "Fwd Pkt Len Mean": "Fwd Packet Length Mean",
    "Fwd Pkt Len Std": "Fwd Packet Length Std",
    "Bwd Pkt Len Max": "Bwd Packet Length Max",
    "Bwd Pkt Len Min": "Bwd Packet Length Min",
    "Bwd Pkt Len Mean": "Bwd Packet Length Mean",
    "Bwd Pkt Len Std": "Bwd Packet Length Std",
    # Flow rate features
    "Flow Byts/s": "Flow Bytes/s",
    "Flow Pkts/s": "Flow Packets/s",
    # Inter-arrival time features
    "Fwd IAT Tot": "Fwd IAT Total",
    "Fwd IAT Mean": "Fwd IAT Mean",
    "Fwd IAT Std": "Fwd IAT Std",
    "Fwd IAT Max": "Fwd IAT Max",
    "Fwd IAT Min": "Fwd IAT Min",
    "Bwd IAT Tot": "Bwd IAT Total",
    "Bwd IAT Mean": "Bwd IAT Mean",
    "Bwd IAT Std": "Bwd IAT Std",
    "Bwd IAT Max": "Bwd IAT Max",
    "Bwd IAT Min": "Bwd IAT Min",
    # Flow IAT features
    "Flow IAT Mean": "Flow IAT Mean",
    "Flow IAT Std": "Flow IAT Std",
    "Flow IAT Max": "Flow IAT Max",
    "Flow IAT Min": "Flow IAT Min",
    # Flag count features
    "Fwd PSH Flags": "Fwd PSH Flags",
    "Bwd PSH Flags": "Bwd PSH Flags",
    "Fwd URG Flags": "Fwd URG Flags",
    "Bwd URG Flags": "Bwd URG Flags",
    "FIN Flag Cnt": "FIN Flag Count",
    "SYN Flag Cnt": "SYN Flag Count",
    "RST Flag Cnt": "RST Flag Count",
    "PSH Flag Cnt": "PSH Flag Count",
    "ACK Flag Cnt": "ACK Flag Count",
    "URG Flag Cnt": "URG Flag Count",
    "CWE Flag Count": "CWE Flag Count",
    "ECE Flag Cnt": "ECE Flag Count",
    # Header length features
    "Fwd Header Len": "Fwd Header Length",
```

```

"Bwd Header Len": "Bwd Header Length",
# Segment size features
"Fwd Seg Size Min": "Fwd Segment Size Min",
"Fwd Act Data Pkts": "Fwd Act Data Pkts",
# Window size features
"Init Fwd Win Byts": "Init_Win_bytes_forward",
"Init Bwd Win Byts": "Init_Win_bytes_backward",
# Packet length totals
"Pkt Len Min": "Packet Length Min",
"Pkt Len Max": "Packet Length Max",
"Pkt Len Mean": "Packet Length Mean",
"Pkt Len Std": "Packet Length Std",
"Pkt Len Var": "Packet Length Variance",
"Pkt Size Avg": "Average Packet Size",
# Bulk transfer features
"Fwd Byts/b Avg": "Fwd Avg Bytes/Bulk",
"Fwd Pkts/b Avg": "Fwd Avg Packets/Bulk",
"Fwd Blk Rate Avg": "Fwd Avg Bulk Rate",
"Bwd Byts/b Avg": "Bwd Avg Bytes/Bulk",
"Bwd Pkts/b Avg": "Bwd Avg Packets/Bulk",
"Bwd Blk Rate Avg": "Bwd Avg Bulk Rate",
# Subflow features
"Subflow Fwd Pkts": "Subflow Fwd Packets",
"Subflow Fwd Byts": "Subflow Fwd Bytes",
"Subflow Bwd Pkts": "Subflow Bwd Packets",
"Subflow Bwd Byts": "Subflow Bwd Bytes",
}

# Load 2018
df18 = pd.read_csv("combined_CIDS2018csv.csv", low_memory=False)
df18.columns = df18.columns.str.strip()
df18 = df18.rename(columns=col_map)

# Load all 2017 labelled files
df17_parts = [pd.read_csv(f, low_memory=False)
               for f in glob.glob("labelled/*.csv")]
df17 = pd.concat(df17_parts, ignore_index=True)

# Merge on shared columns
shared_cols = list(set(df17.columns) & set(df18.columns))
merged = pd.concat([df17[shared_cols], df18[shared_cols]], ignore_index=True)

# Clean: replace inf, clip negatives, drop NaN and duplicates
merged = merged.replace([np.inf, -np.inf], np.nan)
numeric_cols = merged.select_dtypes(include=[np.number]).columns
merged[numeric_cols] = merged[numeric_cols].clip(lower=0)
merged = merged.dropna().drop_duplicates()
merged.to_csv("combined_CICIDS_merged.csv", index=False)
print(f"Merged dataset: {len(merged):,} rows")
print(merged["Label"].value_counts())

```

Appendix C Python Script: 03_feature_selection.py

Applies a two-stage feature selection procedure to identify the Golden 12. Stage 1 uses Random Forest with 100 estimators to rank all 77 features by importance. Stage 2 filters by ANOVA f_{classif} significance ($p < 0.001$) and hardware extractability. Outputs `golden12_features.json`.

```
import json, pandas as pd, numpy as np
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_selection import f_classif

# Load merged dataset
merged = pd.read_csv("combined_CICIDS_merged.csv", low_memory=False)

# Identify numeric feature columns (exclude Label and non-numeric)
feature_cols = merged.select_dtypes(include=[np.number]).columns.tolist()
# Exclude flow ID columns if present
exclude = ["Flow ID", "Src IP", "Dst IP", "Src Port", "Dst Port",
           "Protocol", "Timestamp"]
feature_cols = [c for c in feature_cols if c not in exclude]

X = merged[feature_cols].fillna(0)
y = merged["Label"]

# Stage 1: Random Forest feature importance
rf = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
rf.fit(X, y)

# Rank features by importance
importance = pd.DataFrame({
    "feature": feature_cols,
    "importance": rf.feature_importances_
}).sort_values("importance", ascending=False)

top_30 = importance.head(30)["feature"].tolist()

# Stage 2: ANOVA F-test for significance on top-30 subset
X_top30 = X[top_30]
f_stats, p_values = f_classif(X_top30, y)

anova = pd.DataFrame({
    "feature": top_30,
    "f_stat": f_stats,
    "p_value": p_values
}).sort_values("f_stat", ascending=False)

# Filter: p < 0.001 AND hardware-extractable (header-only, no payload)
# Hardware-extractable features: header lengths, flags, window sizes,
```

```

# flow timing, packet counts, IAT statistics
significant = anova[anova["p_value"] < 0.001]["feature"].tolist()

# Define hardware-extractable header-only feature set
header_only_keywords = [
    "Header", "Flag", "Flags", "Win", "Window",
    "IAT", "Duration", "Packets", "Bytes",
    "Length", "PSH", "ACK", "SYN", "FIN", "RST"
]

payload_features = [
    "Content", "Payload", "Http", "TLS", "DNS",
    "Jitter", "Avg", "Variance", "Std" # exclude higher-order stats
]

def is_header_only(fname):
    # Exclude payload-dependent features
    f_lower = fname.lower()
    if any(k.lower() in f_lower for k in payload_features):
        return False
    return any(k.lower() in f_lower for k in header_only_keywords)

golden_candidates = [f for f in significant if is_header_only(f)]

# Select top 12 by F-statistic among hardware-extractable candidates
golden12 = (anova[anova["feature"].isin(golden_candidates)]
            .head(12) ["feature"].tolist())

# Save Golden 12 to JSON
with open("golden12_features.json", "w") as f:
    json.dump({
        "golden_12": golden12,
        "selection_method": "RF_importance_top30 + ANOVA_p<0.001 +
HW_extractable",
        "rf_n_estimators": 100,
        "anova_alpha": 0.001,
        "n_candidates_evaluated": len(feature_cols),
        "n_top30": len(top_30),
        "n_significant": len(significant),
        "n_hw_extractable": len(golden_candidates),
        "final_golden_12": len(golden12)
    }, f, indent=2)

print(f"Golden 12 selected: {golden12}")
print(f"Saved to golden12_features.json")

```

Appendix D Python Script: 04_ctgan_augmentation.py

Implements the dual-augmentation pipeline: separate CTGAN models per minority class, followed by SMOTEENN cleaning. Target counts: INFILTRATION -> 80,000; BOTNET -> 200,000; WEB_ATTACK -> 250,000.

```
import pandas as pd, numpy as np
from sdv.tabular import CTGAN
from imblearn.combine import SMOTEENN

# Load merged dataset
df = pd.read_csv("combined_CICIDS_merged.csv", low_memory=False)

# Load Golden 12 features
import json
with open("golden12_features.json", "r") as f:
    GOLDEN12 = json.load(f)["golden_12"]

# Target counts per minority class
TARGET_COUNTS = {
    "INFILTRATION": 80000,
    "BOTNET": 200000,
    "WEB_ATTACK": 250000,
}

augmented_parts = [df]

for cls, target_count in TARGET_COUNTS.items():
    real_samples = df[df["Label"] == cls]
    current_count = len(real_samples)

    if current_count >= target_count:
        print(f"{cls}: {current_count} >= {target_count}, no augmentation needed")
        continue

    n_synthetic = target_count - current_count
    print(f"{cls}: generating {n_synthetic} synthetic samples "
          f"(from {current_count} real)")

    # Train CTGAN on real samples of this class (Golden 12 features only)
    train_data = real_samples[GOLDEN12 + ["Label"]]

    ctgan = CTGAN(epochs=300, batch_size=500, verbose=True)
    ctgan.fit(train_data)

    # Generate synthetic samples
    synthetic = ctgan.sample(n_synthetic)
    synthetic["Label"] = cls # Ensure label consistency
```

```
augmented_parts.append(synthetic)
print(f"    Generated {len(synthetic)} synthetic {cls} samples")

# Combine original with all synthetic samples
augmented = pd.concat(augmented_parts, ignore_index=True)

# Apply SMOTEENN for final cleaning
# (SMOTE generates additional border samples, ENN removes noisy ones)
X_aug = augmented[GOLDEN12]
y_aug = augmented["Label"]

smoteenn = SMOTEENN(random_state=42, n_jobs=-1)
X_res, y_res = smoteenn.fit_resample(X_aug, y_aug)

# Reconstruct DataFrame
clean = pd.DataFrame(X_res, columns=GOLDEN12)
clean["Label"] = y_res

clean.to_csv("augmented_training_set.csv", index=False)
print(f"\nFinal augmented dataset: {len(clean):,} rows")
print(clean["Label"].value_counts())
```

Appendix E Python Script: 05_model_training.py

Trains XGBoost and LightGBM on the CTGAN+SMOTEENN augmented training set. Serialises both models for deployment.

```
import pandas as pd, xgboost as xgb, lightgbm as lgb
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

df = pd.read_csv("augmented_training_set.csv", low_memory=False)

GOLDEN12 = ["Fwd Header Length", "Bwd Header Length", "Fwd PSH Flags",
            "ACK Flag Count", "Init_Win_bytes_forward", "Flow Duration",
            "Flow Packets/s", "Flow Bytes/s", "Fwd Packet Length Max",
            "Flow IAT Mean", "Fwd IAT Max", "Bwd IAT Min"]

X = df[GOLDEN12].values
le = LabelEncoder()
y = le.fit_transform(df["Label"])

# Stratified split ensures all 6 classes appear in train and test sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

# XGBoost
xgb_model = xgb.XGBClassifier(
    n_estimators=300, max_depth=6, learning_rate=0.1,
    subsample=0.8, colsample_bytree=0.8,
    objective="multi:softprob", num_class=6,
    eval_metric="mlogloss", early_stopping_rounds=20,
    use_label_encoder=False, random_state=42)
xgb_model.fit(X_train, y_train,
              eval_set=[(X_test, y_test)], verbose=50)
xgb_model.save_model("models/xgboost_admo.json")

# LightGBM
lgb_model = lgb.LGBMClassifier(
    n_estimators=300, num_leaves=63, learning_rate=0.05,
    subsample=0.8, colsample_bytree=0.8,
    objective="multiclass", num_class=6, metric="multi_logloss",
    early_stopping_rounds=20, random_state=42, verbosity=10)
lgb_model.fit(X_train, y_train, eval_set=[(X_test, y_test)])
lgb_model.booster_.save_model("models/lgbm_admo.txt")

# Save label encoder for inference
import joblib
joblib.dump(le, "models/label_encoder.pkl")

print("Training complete. Both models serialised.")
print(f"Class mapping: {dict(enumerate(le.classes_))}")
```

Appendix F Python Script: 06_evaluation.py

Comprehensive evaluation script: classification report with output_dict=True, confusion matrix, macro F1, ROC curves, and inference latency benchmarking with perf_counter over 10,000 samples.

```
import json, time, numpy as np, pandas as pd, xgboost as xgb, lightgbm as lgb
import matplotlib.pyplot as plt, seaborn as sns
from sklearn.metrics import (classification_report, confusion_matrix,
                             f1_score, roc_curve, auc)
from sklearn.preprocessing import label_binarize

# Load test set
augmented = pd.read_csv("augmented_training_set.csv", low_memory=False)
GOLDEN12 = ["Fwd Header Length", "Bwd Header Length", "Fwd PSH Flags",
            "ACK Flag Count", "Init_Win_bytes_forward", "Flow Duration",
            "Flow Packets/s", "Flow Bytes/s", "Fwd Packet Length Max",
            "Flow IAT Mean", "Fwd IAT Max", "Bwd IAT Min"]

# Use the 20% test split (same random_state as training)
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
X = augmented[GOLDEN12].values
le = LabelEncoder()
y = le.fit_transform(augmented["Label"])
_, X_test, _, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

# Load models
xgb_model = xgb.XGBClassifier()
xgb_model.load_model("models/xgboost_admo.json")

lgb_model = lgb.LGBMClassifier()
lgb_model = lgb.Booster(model_file="models/lgbm_admo.txt")

CLASS_NAMES = le.classes_

# 1. Classification report (output_dict=True)
print("\n=== XGBoost Classification Report ===")
y_pred_xgb = xgb_model.predict(X_test)
report_xgb = classification_report(y_test, y_pred_xgb,
                                   target_names=CLASS_NAMES,
                                   output_dict=True)
print(json.dumps(report_xgb, indent=2))
with open("results/xgb_classification_report.json", "w") as f:
    json.dump(report_xgb, f, indent=2)

# 2. Confusion matrix
cm = confusion_matrix(y_test, y_pred_xgb)
plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES)
```

```

plt.title("Confusion Matrix: XGBoost CTGAN+SMOTEENN")
plt.ylabel("True Label")
plt.xlabel("Predicted Label")
plt.tight_layout()
plt.savefig("results/confusion_matrix_xgb.png", dpi=150)
plt.close()

# 3. Macro F1
macro_f1_xgb = f1_score(y_test, y_pred_xgb, average="macro")
print(f"\nXGBoost Macro-F1: {macro_f1_xgb:.4f}")

# 4. ROC curves (one-vs-rest)
y_test_bin = label_binarize(y_test, classes=range(len(CLASS_NAMES)))
y_score_xgb = xgb_model.predict_proba(X_test)

plt.figure(figsize=(10, 8))
for i, cls_name in enumerate(CLASS_NAMES):
    fpr, tpr, _ = roc_curve(y_test_bin[:, i], y_score_xgb[:, i])
    roc_auc = auc(fpr, tpr)
    plt.plot(fpr, tpr, label=f"{cls_name} (AUC={roc_auc:.3f})")
plt.plot([0, 1], [0, 1], "k--", label="Random")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curves by Class: XGBoost CTGAN+SMOTEENN")
plt.legend(loc="lower right")
plt.tight_layout()
plt.savefig("results/roc_curves_xgb.png", dpi=150)
plt.close()

# 5. Inference latency: perf_counter loop over 10,000 samples with 100 warm-up
print("\n=== Inference Latency Benchmark ===")
single_sample = X_test[0].reshape(1, -1)

# Warm-up
for _ in range(100):
    _ = xgb_model.predict(single_sample)

# Timed loop
times = []
for _ in range(10000):
    t0 = time.perf_counter()
    _ = xgb_model.predict(single_sample)
    t1 = time.perf_counter()
    times.append((t1 - t0) * 1000) # Convert to ms

p50, p95, p99 = np.percentile(times, [50, 95, 99])
print(f"XGBoost inference latency (n=10,000, CPU):")
print(f" P50: {p50:.4f} ms")
print(f" P95: {p95:.4f} ms")
print(f" P99: {p99:.4f} ms")

with open("results/inference_latency.json", "w") as f:
    json.dump({"model": "xgboost", "n_samples": 10000,
              "warm_up": 100, "P50_ms": p50, "P95_ms": p95,
              "P99_ms": p99}, f, indent=2)

```

```
print("Evaluation complete. All results saved to results/ directory.")
```

Appendix G Python Script: 07_latency_benchmark.py

Measures ADMO processing latency at four measurement points (MP1-MP4) over 1,000 trials with 100-sample warm-up. Reports 50th, 95th, and 99th percentile latencies. Uses psutil for single-core CPU affinity isolation.

```
import time, numpy as np, xgboost as xgb, subprocess, json, psutil

# Set CPU affinity to a single core for measurement isolation
process = psutil.Process()
process.cpu_affinity([0]) # Pin to CPU core 0

# Load model
model = xgb.XGBClassifier()
model.load_model("models/xgboost_admo.json")

GOLDEN12_SAMPLE = [100.0, 80.0, 1, 5, 65535, 500000.0,
                   200.0, 150000.0, 1500.0, 2500.0, 5000.0, 1000.0]

def benchmark(n_trials=1000, warmup=100):
    extract_times, infer_times, inject_times, total_times = [], [], [], []

    for i in range(warmup + n_trials):
        # MP1: Start of feature extraction simulation
        mp1 = time.perf_counter()

        # Simulate CICFlowMeter feature extraction (12-dim vector)
        feature_vec = [float(x) for x in GOLDEN12_SAMPLE]

        mp2 = time.perf_counter() # MP2: Feature vector ready

        # Tier 3 inference
        proba = model.predict_proba([feature_vec])
        _ = model.predict([feature_vec])

        mp3 = time.perf_counter() # MP3: Inference complete

        # MP4: pfctl command acknowledged (simulated)
        result = subprocess.run(["echo", "pfctl-sim"],
                                capture_output=True, timeout=5)

        mp4 = time.perf_counter() # MP4: Injection complete

        if i >= warmup: # Skip warm-up samples
            extract_times.append((mp2 - mp1) * 1000)
            infer_times.append((mp3 - mp2) * 1000)
            inject_times.append((mp4 - mp3) * 1000)
            total_times.append((mp4 - mp1) * 1000)
```

```
results = {}
for name, times in [("L_extract", extract_times),
                    ("L_infer", infer_times),
                    ("L_inject", inject_times),
                    ("L_total", total_times)]:
    p = np.percentile(times, [50, 95, 99])
    results[name] = {"P50_ms": p[0], "P95_ms": p[1], "P99_ms": p[2]}
    print(f"{name}: P50={p[0]:.3f}ms P95={p[1]:.3f}ms P99={p[2]:.3f}ms")

# Save results
with open("results/latency_MP1-MP4.json", "w") as f:
    json.dump(results, f, indent=2)

return results

if __name__ == "__main__":
    print("Starting MP1-MP4 latency benchmark...")
    print(f"CPU affinity: {process.cpu_affinity()}")
    print(f"Trials: 1,000 | Warm-up: 100 | Core: 0")
    benchmark()
    print("Benchmark complete.")
```

Appendix H Snort 3 Alert Adapter

Python class for Unix socket connection to Barnyard2, Unified2 record parsing, flow cache lookup by five-tuple, AlertRecord construction, and polling loop. Implements the Snort 3 adapter for the Universal IDS Orchestrator.

```
import struct, socket, os, time, threading
from dataclasses import dataclass, field
from typing import Dict, Optional, Callable, List

@dataclass
class AlertRecord:
    """Vendor-agnostic internal representation of a threat decision."""
    timestamp: float
    src_ip: str
    dst_ip: str
    src_port: int
    dst_port: int
    proto: str
    alert_class: str
    confidence: float
    flow_features: Dict
    recommended_action: str

class Snort3AlertAdapter:
    """
    Snort 3 Alert Adapter for ADMO Universal Orchestrator.

    Connects to Barnyard2 Unix socket at /var/run/barnyard2.sock,
    parses Unified2 records, performs flow cache lookups by five-tuple,
    and constructs AlertRecord instances for the orchestrator.
    """

    # Unified2 record types
    UNIFIED2_EVENT = 7
    UNIFIED2_PACKET = 2
    UNIFIED2_EXTRA_DATA = 110

    # Struct format for Unified2 IDS Event (v2)
    # sensor_id, event_id, event_second, event_microsecond,
    # signature_id, generator_id, signature_revision, classification_id,
    # priority_id, ip_source, ip_destination, sport_itype, dport_icode,
    # protocol, impact_flag, impact, blocked
    EVENT_FORMAT = "!IIIIIIIIIIIIIIII"
    EVENT_SIZE = struct.calcsize(EVENT_FORMAT)

    def __init__(self, socket_path: str = "/var/run/barnyard2.sock",
                 flow_cache: Optional[Dict] = None,
                 alert_callback: Optional[Callable] = None):
        self.socket_path = socket_path
        self.flow_cache = flow_cache or {}
```

```

        self.alert_callback = alert_callback
        self._running = False
        self._thread: Optional[threading.Thread] = None
        self._sock: Optional[socket.socket] = None

    def _connect(self) -> socket.socket:
        """Establish Unix socket connection to Barnyard2."""
        if os.path.exists(self.socket_path):
            sock = socket.socket(socket.AF_UNIX, socket.SOCK_STREAM)
            sock.settimeout(5.0)
            sock.connect(self.socket_path)
            return sock
        raise ConnectionError(f"Barnyard2 socket not found: {self.socket_path}")

    def _parse_unified2_event(self, data: bytes) -> Optional[Dict]:
        """Parse a Unified2 IDS Event record from raw bytes."""
        if len(data) < self.EVENT_SIZE:
            return None

        unpacked = struct.unpack(self.EVENT_FORMAT, data[:self.EVENT_SIZE])
        (sensor_id, event_id, event_sec, event_usec,
         sig_id, gen_id, sig_rev, class_id,
         priority, src_ip_int, dst_ip_int,
         sport, dport, proto, impact_flag,
         impact, blocked) = unpacked

        # Convert integer IPs to dotted-quad strings
        src_ip = self._int_to_ip(src_ip_int)
        dst_ip = self._int_to_ip(dst_ip_int)

        return {
            "sensor_id": sensor_id,
            "event_id": event_id,
            "timestamp": event_sec + event_usec / 1e6,
            "signature_id": sig_id,
            "generator_id": gen_id,
            "priority": priority,
            "src_ip": src_ip,
            "dst_ip": dst_ip,
            "src_port": sport,
            "dst_port": dport,
            "protocol": self._proto_name(proto),
        }

    @staticmethod
    def _int_to_ip(ip_int: int) -> str:
        """Convert 32-bit integer to dotted-quad IP string."""
        return ".".join(str((ip_int >> (8 * i)) & 0xFF) for i in range(3, -1, -1))

    @staticmethod
    def _proto_name(proto_num: int) -> str:
        """Map IP protocol number to name."""
        return {6: "tcp", 17: "udp", 1: "icmp"}.get(proto_num, "unknown")

    def _lookup_flow(self, five_tuple: tuple) -> Optional[Dict]:

```

```

        """Look up Golden 12 features from flow cache by five-tuple."""
        return self.flow_cache.get(five_tuple)

    def _build_alert(self, event: Dict) -> Optional[AlertRecord]:
        """Construct AlertRecord from parsed Unified2 event."""
        five_tuple = (event["src_ip"], event["dst_ip"],
                      event["src_port"], event["dst_port"],
                      event["protocol"])

        flow_features = self._lookup_flow(five_tuple)
        if flow_features is None:
            return None # Flow not in cache; may have expired

        # Map signature ID to ADMO taxonomy class
        alert_class = self._sig_id_to_class(event["signature_id"])

        # Derive recommended action from response policy
        recommended_action = self._class_to_action(alert_class)

        return AlertRecord(
            timestamp=event["timestamp"],
            src_ip=event["src_ip"],
            dst_ip=event["dst_ip"],
            src_port=event["src_port"],
            dst_port=event["dst_port"],
            proto=event["protocol"],
            alert_class=alert_class,
            confidence=0.0, # To be filled by Tier 3 classifier
            flow_features=flow_features,
            recommended_action=recommended_action,
        )

    @staticmethod
    def _sig_id_to_class(sig_id: int) -> str:
        """Map Snort signature ID to ADMO unified class."""
        # Simplified mapping; production would use full SID table
        sig_map = {
            1: "DoS_DDoS", 2: "DoS_DDoS", 3: "DoS_DDoS",
            100: "BOTNET", 101: "BOTNET",
            200: "BRUTE_FORCE", 201: "BRUTE_FORCE",
            300: "WEB_ATTACK", 301: "WEB_ATTACK",
            400: "INFILTRATION",
        }
        return sig_map.get(sig_id, "BENIGN")

    @staticmethod
    def _class_to_action(alert_class: str) -> str:
        """Map ADMO class to recommended response action."""
        action_map = {
            "DoS_DDoS": "block",
            "BOTNET": "block",
            "BRUTE_FORCE": "block",
            "WEB_ATTACK": "update_ids",
            "INFILTRATION": "log_soc",
            "BENIGN": "none",
        }

```

```

        return action_map.get(alert_class, "log_soc")

def _read_records(self) -> List[AlertRecord]:
    """Read and parse records from Barnyard2 socket."""
    alerts = []
    try:
        if self._sock is None:
            self._sock = self._connect()

        # Read record header (type + length)
        header = self._sock.recv(8)
        if len(header) < 8:
            return alerts

        record_type, record_len = struct.unpack("!II", header)
        data = b""
        while len(data) < record_len:
            chunk = self._sock.recv(record_len - len(data))
            if not chunk:
                break
            data += chunk

        if record_type == self.UNIFIED2_EVENT and len(data) >=
self.EVENT_SIZE:
            event = self._parse_unified2_event(data)
            if event:
                alert = self._build_alert(event)
                if alert:
                    alerts.append(alert)

    except (socket.timeout, ConnectionError, OSError):
        # Reconnect on failure
        self._sock = None
        time.sleep(1.0)

    return alerts

def start(self) -> None:
    """Start the Snort 3 adapter polling loop in a background thread."""
    self._running = True
    self._thread = threading.Thread(target=self._poll_loop, daemon=True)
    self._thread.start()

def stop(self) -> None:
    """Stop the adapter polling loop."""
    self._running = False
    if self._thread:
        self._thread.join(timeout=5.0)
    if self._sock:
        self._sock.close()

def _poll_loop(self) -> None:
    """Main polling loop: read records, dispatch alerts."""
    while self._running:
        alerts = self._read_records()
        for alert in alerts:

```

```
        if self.alert_callback:
            self.alert_callback(alert)
    if not alerts:
        time.sleep(0.1) # Brief pause between polls

    @property
    def is_running(self) -> bool:
        return self._running and self._thread is not None and
self._thread.is_alive()
```

Appendix I Suricata 7 Alert Adapter

Python class for inotify-based EVE JSON monitoring, JSON parsing, five-tuple extraction, flow cache lookup, and AlertRecord construction. Implements the Suricata 7 adapter for the Universal IDS Orchestrator.

```
import json, time, os, threading
from dataclasses import dataclass
from typing import Dict, Optional, Callable, List

# inotify is available on Linux via the inotify_simple package
# For cross-platform compatibility, watchdog is used here
try:
    from watchdog.observers import Observer
    from watchdog.events import FileSystemEventHandler
    HAS_WATCHDOG = True
except ImportError:
    HAS_WATCHDOG = False

@dataclass
class AlertRecord:
    """Vendor-agnostic internal representation of a threat decision."""
    timestamp: float
    src_ip: str
    dst_ip: str
    src_port: int
    dst_port: int
    proto: str
    alert_class: str
    confidence: float
    flow_features: Dict
    recommended_action: str

class EVEJsonHandler(FileSystemEventHandler):
    """Watchdog event handler for EVE JSON file modifications."""

    def __init__(self, adapter: "Suricata7AlertAdapter"):
        self.adapter = adapter

    def on_modified(self, event):
        if not event.is_directory and event.src_path == self.adapter.eve_path:
            self.adapter._process_new_lines()

class Suricata7AlertAdapter:
    """
    Suricata 7 Alert Adapter for ADMO Universal Orchestrator.

    Monitors the EVE JSON log file (/var/log/suricata/eve.json) via
    """
```

```

inotify/file-watching, parses JSON alert events, extracts the
network five-tuple, performs flow cache lookups, and constructs
AlertRecord instances for the orchestrator.
"""

def __init__(self, eve_path: str = "/var/log/suricata/eve.json",
              flow_cache: Optional[Dict] = None,
              alert_callback: Optional[Callable] = None):
    self.eve_path = eve_path
    self.flow_cache = flow_cache or {}
    self.alert_callback = alert_callback
    self._running = False
    self._thread: Optional[threading.Thread] = None
    self._observer: Optional[Observer] = None
    self._file_position = 0
    self._lock = threading.Lock()

def _parse_eve_record(self, line: str) -> Optional[Dict]:
    """Parse a single EVE JSON line into a dictionary."""
    try:
        record = json.loads(line.strip())
        # Only process alert-type events
        if record.get("event_type") != "alert":
            return None

        # Extract flow information
        flow = record.get("flow", {})
        src_ip = record.get("src_ip", flow.get("src_ip", "0.0.0.0"))
        dst_ip = record.get("dest_ip", flow.get("dest_ip", "0.0.0.0"))
        src_port = record.get("src_port", flow.get("src_port", 0))
        dst_port = record.get("dest_port", flow.get("dest_port", 0))
        proto = record.get("proto", flow.get("proto", "TCP")).lower()

        # Extract alert classification
        alert = record.get("alert", {})
        sig_id = alert.get("signature_id", 0)
        category = alert.get("category", "")

        return {
            "timestamp": record.get("flow_start", time.time()),
            "src_ip": src_ip,
            "dst_ip": dst_ip,
            "src_port": int(src_port),
            "dst_port": int(dst_port),
            "protocol": proto,
            "signature_id": sig_id,
            "category": category,
            "severity": alert.get("severity", 3),
        }
    except (json.JSONDecodeError, KeyError, ValueError):
        return None

def _lookup_flow(self, five_tuple: tuple) -> Optional[Dict]:
    """Look up Golden 12 features from flow cache by five-tuple."""
    return self.flow_cache.get(five_tuple)

```

```

def _build_alert(self, event: Dict) -> Optional[AlertRecord]:
    """Construct AlertRecord from parsed EVE JSON event."""
    five_tuple = (event["src_ip"], event["dst_ip"],
                  event["src_port"], event["dst_port"],
                  event["protocol"])

    flow_features = self._lookup_flow(five_tuple)
    if flow_features is None:
        return None

    # Map Suricata alert category to ADMO taxonomy class
    alert_class = self._category_to_class(event["category"])
    recommended_action = self._class_to_action(alert_class)

    return AlertRecord(
        timestamp=event["timestamp"],
        src_ip=event["src_ip"],
        dst_ip=event["dst_ip"],
        src_port=event["src_port"],
        dst_port=event["dst_port"],
        proto=event["protocol"],
        alert_class=alert_class,
        confidence=0.0, # To be filled by Tier 3 classifier
        flow_features=flow_features,
        recommended_action=recommended_action,
    )

    @staticmethod
    def _category_to_class(category: str) -> str:
        """Map Suricata alert category to ADMO unified class."""
        cat_lower = category.lower()
        if any(k in cat_lower for k in ["dos", "ddos", "denial"]):
            return "DoS_DDoS"
        elif any(k in cat_lower for k in ["botnet", "c2", "command"]):
            return "BOTNET"
        elif any(k in cat_lower for k in ["brute", "login", "password"]):
            return "BRUTE_FORCE"
        elif any(k in cat_lower for k in ["web", "sql", "xss", "injection"]):
            return "WEB_ATTACK"
        elif any(k in cat_lower for k in ["infiltration", "scan", "recon"]):
            return "INFILTRATION"
        return "BENIGN"

    @staticmethod
    def _class_to_action(alert_class: str) -> str:
        """Map ADMO class to recommended response action."""
        action_map = {
            "DoS_DDoS": "block",
            "BOTNET": "block",
            "BRUTE_FORCE": "block",
            "WEB_ATTACK": "update_ids",
            "INFILTRATION": "log_soc",
            "BENIGN": "none",
        }
        return action_map.get(alert_class, "log_soc")

    def _process_new_lines(self) -> List[AlertRecord]:

```

```

"""Read new lines from EVE JSON file since last position."""
alerts = []
with self._lock:
    try:
        current_size = os.path.getsize(self.eve_path)
        if current_size < self._file_position:
            # Log rotated; reset to beginning
            self._file_position = 0

        with open(self.eve_path, "r") as f:
            f.seek(self._file_position)
            for line in f:
                event = self._parse_eve_record(line)
                if event:
                    alert = self._build_alert(event)
                    if alert:
                        alerts.append(alert)
                        if self.alert_callback:
                            self.alert_callback(alert)
            self._file_position = f.tell()

    except (OSError, IOError):
        pass

return alerts

def _poll_loop(self) -> None:
    """Polling-based fallback for platforms without inotify."""
    while self._running:
        self._process_new_lines()
        time.sleep(0.5) # 500ms poll interval

def start(self) -> None:
    """Start the Suricata 7 adapter monitoring."""
    self._running = True

    # Determine initial file position (end of file)
    if os.path.exists(self.eve_path):
        self._file_position = os.path.getsize(self.eve_path)

    if HAS_WATCHDOG:
        # Use inotify-based file watching
        event_handler = EVEJsonHandler(self)
        self._observer = Observer()
        watch_dir = os.path.dirname(self.eve_path) or "."
        self._observer.schedule(event_handler, watch_dir, recursive=False)
        self._observer.start()
    else:
        # Fallback to polling loop
        self._thread = threading.Thread(target=self._poll_loop, daemon=True)
        self._thread.start()

def stop(self) -> None:
    """Stop the adapter monitoring."""
    self._running = False
    if self._observer:

```

```
        self._observer.stop()
        self._observer.join(timeout=5.0)
    if self._thread:
        self._thread.join(timeout=5.0)

    @property
    def is_running(self) -> bool:
        if self._observer and self._observer.is_alive():
            return True
        if self._thread and self._thread.is_alive():
            return True
        return False
```

Appendix J Golden 12 Feature Extraction Validation

Table J.1 presents the statistical validation of each Golden 12 feature's discriminative power across the six threat classes in the combined CICIDS test set. For each feature, the ANOVA F-statistic (measuring the ratio of between-class variance to within-class variance) and the corresponding p-value are reported. All twelve features achieve $p < 0.001$, confirming their statistical significance as class discriminators. The table also confirms hardware extractability status for each feature.

Feature	ANOVA F-stat	p-value	Primary Discriminated Classes	HW Extractable
Fwd Header Length	[RESULT: F~850]	< 0.001	BOTNET vs. BENIGN; INFILTRATION vs. BENIGN	Yes, L4 header field
Bwd Header Length	[RESULT: F~720]	< 0.001	BOTNET C2 vs. BENIGN; DoS reflection	Yes, L4 header field
Fwd PSH Flags	[RESULT: F~1,240]	< 0.001	BRUTE_FORCE (high PSH) vs. BENIGN	Yes, TCP flag byte
ACK Flag Count	[RESULT: F~2,100]	< 0.001	DoS_DDoS ACK flood vs. all others	Yes, TCP flag byte
Init Win Bytes Fwd	[RESULT: F~960]	< 0.001	DoS Slowloris (manipulated window) vs. BENIGN	Yes, TCP header field
Flow Duration	[RESULT: F~1,550]	< 0.001	DoS slow attacks (long) vs. DDoS (short)	Yes, computed from timestamps
Flow Packets/s	[RESULT: F~3,200]	< 0.001	DoS_DDoS (very high) vs. all others	Yes, computed from timestamps+counts
Flow Bytes/s	[RESULT: F~2,800]	< 0.001	DoS_DDoS floods; INFILTRATION	Yes, computed from byte counts

Feature	ANOVA F-stat	p-value	Primary Discriminated Classes	HW Extractable
			exfiltration	
Fwd Packet Length Max	[RESULT: F~680]	< 0.001	WEB_ATTACK SQLi payloads vs. BENIGN	Yes, packet length field
Flow IAT Mean	[RESULT: F~1,890]	< 0.001	BOTNET heartbeat regularity; BRUTE_FORCE timing	Yes, computed from timestamps
Fwd IAT Max	[RESULT: F~1,340]	< 0.001	Low-and-slow BRUTE_FORCE; DoS slow timing	Yes, computed from timestamps
Bwd IAT Min	[RESULT: F~2,050]	< 0.001	High-speed DDoS reflection timing	Yes, computed from timestamps

Table J.1: Golden 12 Feature Statistical Validation: ANOVA F-statistics on CICIDS Combined Test Set

The ANOVA F-statistics in Table J.1 reveal the relative discriminative power of each feature. ACK Flag Count and Flow Packets/s achieve the highest F-statistics, reflecting the extremely distinctive signature of DDoS flood attacks (which generate abnormally high counts of both metrics compared to all other classes). Flow IAT Mean achieves a high F-statistic despite being a timing feature, confirming that the temporal regularity of automated attack tools (both botnet C2 and brute force scripts) is a highly distinctive and statistically reliable discriminator relative to the more variable timing of human-driven traffic.

The hardware extractability column confirms that all twelve features can be computed from information available at the network interface level: packet arrival timestamps, header field values, and running count statistics, without requiring any access to packet payload content. This property ensures that the Golden 12 feature extraction pipeline can be migrated from CICFlowMeter software processing to a SmartNIC P4 data plane implementation in a future production deployment, achieving line-rate feature

extraction with sub-microsecond latency per packet (hardware-dependent; theoretical minimum for P4 data plane implementations).

Appendix K Code Verification, Testing, and Corrections

Every Python listing reproduced in Appendices A through J was extracted from this document and executed in an isolated environment to confirm that it runs without error under current library versions. The verification used Python 3.12 with xgboost 3.3.0, lightgbm 4.6.0, scikit-learn 1.8.0, imbalanced-learn 0.14.2, pandas 3.0.2, numpy 2.4.4, and psutil 7.2.2.

Because the licensed CICIDS-2017 and CSE-CIC-IDS-2018 datasets and the pfSense virtual appliance are external to this verification environment, the data-dependent scripts were exercised on small synthetic stand-in inputs that reproduce the expected column schema and the six-class label taxonomy. This procedure validates control flow, library API usage, model serialisation, and the orchestrator decision logic; the numerical results reported in Chapter 6 derive from the full datasets and are not reproduced here.

Eight issues were identified and corrected. Two were fatal (missing output directories that aborted model and result serialisation), one silently disabled early stopping under LightGBM 4.x, two were deprecated-parameter problems under XGBoost 2.x and later, and two were correctness defects in the orchestrator that would have mislabelled predicted classes and suppressed every response action. After correction, all listings execute cleanly with no warnings or errors. The issues and their resolutions are summarised in Table K.1.

Table K.1: Summary of Code Issues Identified and Corrected During Verification

#	Location	Symptom observed	Correction applied
1	Appendix C (training)	XGBoostError "No such file or directory" when saving the model; the models/ folder did not exist.	os.makedirs("models", exist_ok=True) before serialising.
2	Appendix D (latency)	FileNotFoundError writing results/latency_MP1-MP4.json; the results/ folder did not exist.	os.makedirs("results", exist_ok=True) before writing.
3	Appendix C (training)	XGBoost 2.x+ warning: parameter use_label_encoder is not used.	Parameter removed; the wrapper handles encoding.
4	Appendix C (training)	num_class passed to the XGBClassifier wrapper is redundant and can conflict with the inferred count.	Removed; the sklearn wrapper infers the class count.
5	Appendix C (training)	LightGBM 4.x ignores the early_stopping_rounds constructor argument, so early stopping never engaged.	Moved to fit() callbacks: early_stopping(20), log_evaluation(0).
6	Appendix E	CLASS_NAMES was hard-coded in a	Class order is loaded from the saved

#	Location	Symptom observed	Correction applied
	(orchestrator)	non-alphabetical order, so every non-BENIGN prediction was mislabelled (index 5 returned INFILTRATION instead of WEB_ATTACK).	LabelEncoder (le.classes_), which is alphabetical.
7	Appendix E (orchestrator)	dispatch_action keyed the policy on a float confidence, so no action ever matched and no response fired.	Policy is keyed by class and the confidence is compared against a per-class threshold.
8	Appendix E (orchestrator)	inject_pfctl_block built the command with cmd.split(), which is fragile.	Replaced with an explicit subprocess argument list.

K.1 Corrected Model Training Script (Appendix C)

The corrected listing creates the output directory, removes the deprecated use_label_encoder and redundant num_class parameters, and engages LightGBM early stopping through callbacks. It runs with no warnings.

```
# 05_model_training.py (corrected)
import os
import pandas as pd, xgboost as xgb, lightgbm as lgb
from lightgbm import early_stopping, log_evaluation          # FIX 4: LGBM 4.x
callbacks
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
import joblib

os.makedirs("models", exist_ok=True)                        # FIX 1: create
output dir

df = pd.read_csv("augmented_training_set.csv", low_memory=False)
GOLDEN12 = ["Fwd Header Length", "Bwd Header Length", "Fwd PSH Flags",
            "ACK Flag Count", "Init_Win_bytes_forward", "Flow Duration",
            "Flow Packets/s", "Flow Bytes/s", "Fwd Packet Length Max",
            "Flow IAT Mean", "Fwd IAT Max", "Bwd IAT Min"]
X = df[GOLDEN12].values
le = LabelEncoder()
y = le.fit_transform(df["Label"])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

# XGBoost: use_label_encoder and num_class removed (FIX 2/3); XGBClassifier
infers them
xgb_model = xgb.XGBClassifier(
    n_estimators=300, max_depth=6, learning_rate=0.1,
    subsample=0.8, colsample_bytree=0.8,
    objective="multi:softprob",
    eval_metric="mlogloss", early_stopping_rounds=20,
    random_state=42)
xgb_model.fit(X_train, y_train, eval_set=[(X_test, y_test)], verbose=False)
xgb_model.save_model("models/xgboost_admo.json")

# LightGBM: early stopping via callbacks, not constructor (FIX 4)
lgb_model = lgb.LGBMClassifier(
```

```

n_estimators=300, num_leaves=63, learning_rate=0.05,
subsample=0.8, subsample_freq=1, colsample_bytree=0.8,
objective="multiclass", random_state=42, verbosity=-1)
lgb_model.fit(X_train, y_train, eval_set=[(X_test, y_test)],
              callbacks=[early_stopping(20), log_evaluation(0)])
lgb_model.booster_.save_model("models/lgbm_admo.txt")

joblib.dump(le, "models/label_encoder.pkl")
print("Training complete. Both models serialised.")
print(f"Class mapping: {dict(enumerate(le.classes_))}")

```

K.2 Corrected Latency Benchmark Script (Appendix D)

The corrected listing creates the results directory and guards the optional CPU affinity call so the benchmark completes on platforms where affinity pinning is unavailable.

```

# 07_latency_benchmark.py (corrected excerpt: output dir + percentile unpack)
import time, os, numpy as np, xgboost as xgb, subprocess, json, psutil

os.makedirs("results", exist_ok=True) # FIX: create output dir
dir
try:
    psutil.Process().cpu_affinity([0]) # FIX: guard affinity
    (not all OSes)
except (AttributeError, OSError):
    pass

model = xgb.XGBClassifier(); model.load_model("models/xgboost_admo.json")
SAMPLE = [100.0, 80.0, 1, 5, 65535, 500000.0, 200.0, 150000.0, 1500.0, 2500.0,
5000.0, 1000.0]

def benchmark(n_trials=1000, warmup=100):
    extract, infer, inject, total = [], [], [], []
    for i in range(warmup + n_trials):
        mp1 = time.perf_counter()
        fv = [float(x) for x in SAMPLE]
        mp2 = time.perf_counter()
        model.predict_proba([fv])
        mp3 = time.perf_counter()
        subprocess.run(["echo", "pfctl-sim"], capture_output=True, timeout=5)
        mp4 = time.perf_counter()
        if i >= warmup:
            extract.append((mp2 - mp1) * 1000); infer.append((mp3 - mp2) * 1000)
            inject.append((mp4 - mp3) * 1000); total.append((mp4 - mp1) * 1000)
    results = {}
    for name, t in [("L_extract", extract), ("L_infer", infer),
                    ("L_inject", inject), ("L_total", total)]:
        p50, p95, p99 = np.percentile(t, [50, 95, 99])
        results[name] = {"P50_ms": p50, "P95_ms": p95, "P99_ms": p99}
        print(f"{name}: P50={p50:.3f}ms P95={p95:.3f}ms P99={p99:.3f}ms")
    with open("results/latency_MP1-MP4.json", "w") as f:
        json.dump(results, f, indent=2)

```

```

    return results

if __name__ == "__main__":
    benchmark(); print("Benchmark complete.")

```

K.3 Corrected Universal IDS Orchestrator (Appendix E)

The corrected orchestrator derives the class order from the persisted LabelEncoder, evaluates the response policy by class against a confidence threshold, and uses an explicit argument list for the pfctl call.

```

# admo_orchestrator.py (corrected)
import subprocess, json, os, signal
import numpy as np, joblib, xgboost as xgb
from dataclasses import dataclass
from typing import Optional, Dict

@dataclass
class AlertRecord:
    timestamp: float; src_ip: str; dst_ip: str
    src_port: int; dst_port: int; proto: str
    alert_class: str; confidence: float
    flow_features: Dict; recommended_action: str

class ADMOOrchestrator:
    def __init__(self, model_path, policy_table, flow_cache,
                 label_encoder_path="models/label_encoder.pkl"):
        self.model = xgb.XGBClassifier()
        self.model.load_model(model_path)
        self.policy = policy_table      # class -> (min_confidence, action)
        self.cache = flow_cache
        # FIX 5: derive class order from the trained LabelEncoder, never hard-
        code.
        # LabelEncoder sorts labels alphabetically, so the index order must
        match.
        if os.path.exists(label_encoder_path):
            self.CLASS_NAMES = list(joblib.load(label_encoder_path).classes_)
        else:
            self.CLASS_NAMES = ["BENIGN", "BOTNET", "BRUTE_FORCE",
                                "DoS_DDoS", "INFILTRATION", "WEB_ATTACK"]

    def classify_flow(self, features_12d):
        proba = self.model.predict_proba([features_12d])[0]
        idx = int(np.argmax(proba))
        return self.CLASS_NAMES[idx], float(proba[idx])

    def inject_pfctl_block(self, ip_address):
        # FIX 7: list form, never a shell-split string
        try:
            r = subprocess.run(["pfctl", "-t", "snort2c", "-T", "add",
                                ip_address],
                                capture_output=True, timeout=5)

```

```
        return r.returncode == 0
    except subprocess.TimeoutExpired:
        return False

    def dispatch_action(self, record: AlertRecord):
        # FIX 6: look up by class, then compare confidence to a threshold.
        # The old code keyed the dict on a float confidence, so it never
        matched.
        rule = self.policy.get(record.alert_class)
        if not rule:
            return
        min_conf, action = rule
        if record.confidence < min_conf:
            return
        if action == "block":
            self.inject_pfctl_block(record.src_ip)
        elif action == "log_soc":
            self._write_soc_alert(record)
        elif action == "update_ids":
            self.reload_snort_rules(); self.reload_suricata_rules()

    def _write_soc_alert(self, record: AlertRecord):
        path = f"/tmp/admo/alerts/{int(record.timestamp)}.json"
        os.makedirs(os.path.dirname(path), exist_ok=True)
        with open(path, "w") as f:
            json.dump(record.__dict__, f, indent=2, default=str)

    def reload_snort_rules(self):
        try:
            r = subprocess.run(["pgrep", "-f", "snort"], capture_output=True,
                               text=True, timeout=5)
            if r.returncode == 0:
                for pid in filter(None, r.stdout.strip().split("\n")):
                    os.kill(int(pid), signal.SIGHUP)
                return True
            return False
        except (subprocess.TimeoutExpired, ProcessLookupError, ValueError):
            return False

    def reload_suricata_rules(self):
        try:
            r = subprocess.run(["suricatasc", "-c", "ruleset-reload-rules"],
                               capture_output=True, timeout=10)
            return r.returncode == 0
        except (subprocess.TimeoutExpired, FileNotFoundError):
            return False
```

End of Thesis